

8 Functional description

8.1 Block description

The CRN120 IC consist of EEPROM, SRAM, NFC interface, Digital Control Unit (Command interpreter, Anticollision, Arbiter/Status registers, I²C control and Memory Interface), Power Management and an I²C slave interface. Energy and data are transferred via an antenna consisting of a coil with a few turns, which is coupled to CRN120.

8.2 NFC interface

The passive NFC interface is compliant to the [NFC Forum Type 2 Tag](#) standard.

It requires to be supplied by an NFC field (e.g. NFC enabled device) always to be able to receive appropriate commands and send the related responses.

As defined in NFC Forum for both directions of data communication, there is one start bit (start of frame) at the beginning of each frame. Each byte is transmitted with an odd parity bit at the end. The least significant bit of the byte 0 of the selected block is transmitted first.

For a multi-byte parameter, the least significant byte is always transmitted first. For example, when reading from the memory using the READ command, byte 0 from the addressed block is transmitted first, followed by bytes 1 to byte 3 out of this block. The same sequence continues for the next block and all subsequent blocks.

8.2.1 Data integrity

The following mechanisms are implemented in the contactless communication link between the NFC device and the CRN120 IC to ensure very reliable data transmission:

- 16 bits CRC per block
- Parity bits for each byte
- Bit count checking
- Bit coding to distinguish between "1", "0" and "no information"
- Channel monitoring (protocol sequence and bit stream analysis)

The commands are initiated by the NFC device and controlled by the Digital Control Unit of the CRN120 IC. The command response depends on the state of the IC, and for memory operations, the access conditions valid for the corresponding page.

8.2.2 NFC state machine

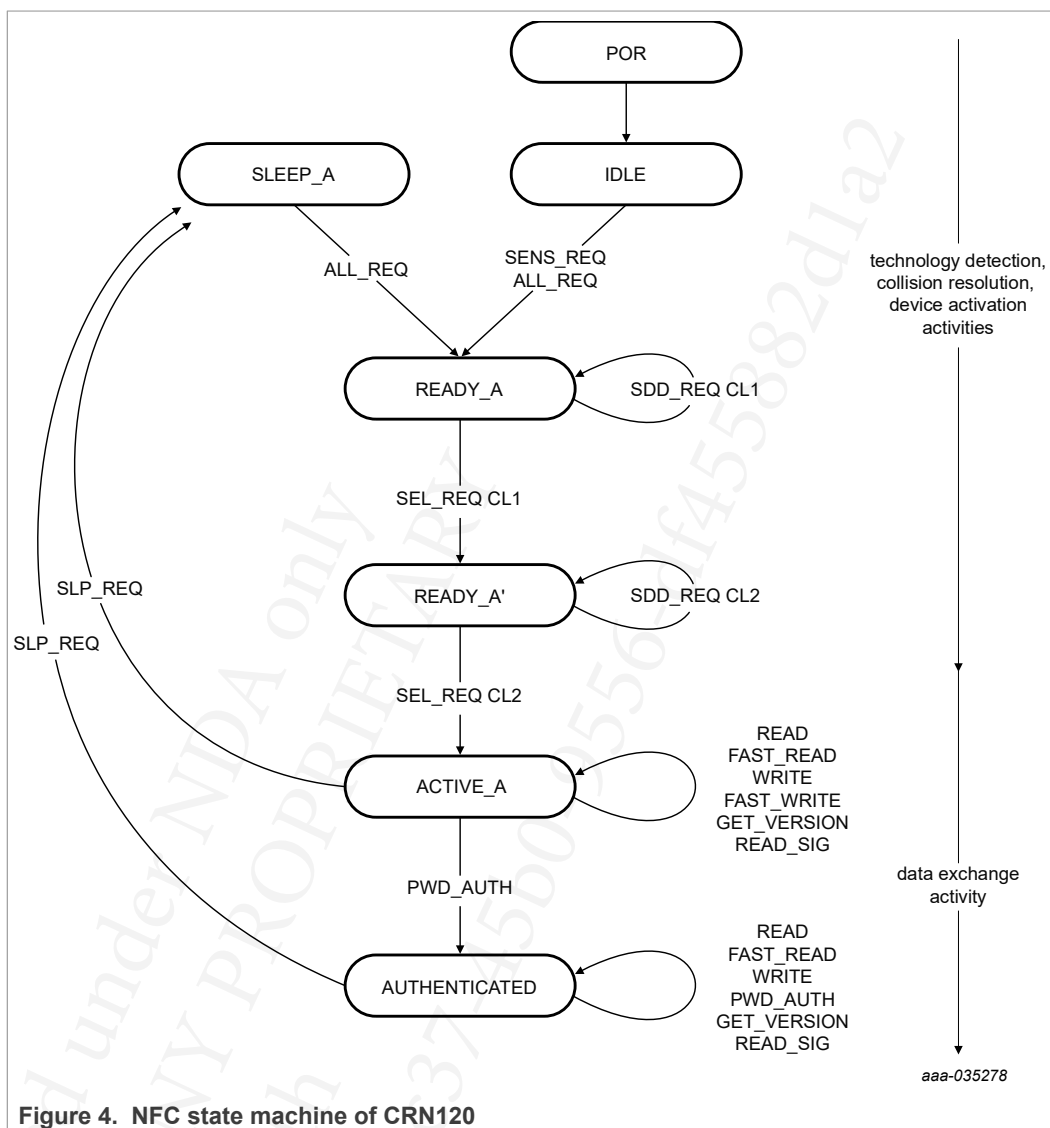


Figure 4. NFC state machine of CRN120

The overall NFC state machine is summarized in [Figure 4](#). When an error is detected or an unexpected command is received, in each state the tag returns to IDLE or SLEEP_A state as defined in [NFC Forum](#).

8.2.2.1 IDLE state

After a Power-On Reset (POR), the CRN120 switches to the default waiting state, namely the IDLE state. It exits IDLE toward READY_A state when a SENS_REQ or a ALL_REQ command is received from the NFC device. Any other data received while in IDLE state is interpreted as an error, and the CRN120 remains in the IDLE state.

8.2.2.2 READY_A state

In the READY_A state, the NFC device resolves the first part of the UID (3 bytes) using the SDD_REQ or SEL_REQ commands for cascade level 1. READY_A state is correctly exited after.

- receiving SEL_REQ command from cascade level 1 with the matching of complete first part of the UID. In this case, the NFC device switches the CRN120 into READY_A' state where the second part of the UID gets resolved.

Remark: The response of the CRN120 to the SEL_REQ command is the SEL_RES byte with cascade bit set to 1b indicating that UID is not complete.

8.2.2.3 READY_A' state

In the READY_A' state, the NFC device resolves the second part of the UID (4 bytes) using the SDD_REQ or SEL_REQ command for cascade level 2. READY_A' state is correctly exited after.

- receiving SEL_REQ command from cascade level 2 with the matching of complete second part of the UID. In this case, the NFC device switches the CRN120 into ACTIVE_A state where all application-related commands can be executed.

Remark: The response of the CRN120 to the SEL_REQ command in READY_A' state is the SEL_RES byte with cascade bit cleared to indicate, that CRN120 is now uniquely selected and only this device will communicate with the NFC device even when other contactless devices are present in the NFC device field.

8.2.2.4 ACTIVE_A state

All unprotected memory operations are operated in the ACTIVE_A and AUTHENTICATED states.

The ACTIVE_A state is exited with the PWD_AUTH command or with the SLP_REQ command.

Upon reception of a correct password within PWD_AUTH command, the CRN120 transits to AUTHENTICATED state after responding with PACK.

With the SLP_REQ command, the CRN120 transits to the SLEEP_A state.

Any other invalid command in ACTIVE_A state is interpreted as an error. Depending on its previous state, the CRN120 returns to either to the IDLE or SLEEP_A state.

8.2.2.5 AUTHENTICATED state

Protected memory operations are only operated in the AUTHENTICATED state, however access to the unprotected memory is possible, too.

The AUTHENTICATED state is exited with the HLTA command and upon reception, the CRN120 transits to the HALT state.

Any other invalid command in AUTHENTICATED state is interpreted as an error. Depending on its previous state, the CRN120 returns to either to the IDLE or HALT state.

8.2.2.6 SLEEP_A state

SLEEP_A and IDLE states constitute the two waiting states implemented in the CRN120. An already processed CRN120 in ACTIVE_A or AUTHENTICATED state can be set into the SLEEP_A state using the SLP_REQ command. In the anti-collision phase, this state helps the NFC device distinguish between processed tags and tags yet to be selected. The CRN120 can only exit SLEEP_A state upon execution of the ALL_REQ command. Any other data received when the device is in this state is interpreted as an error, and CRN120 state remains unchanged.

8.3 Memory organization

The memory map is detailed in [Table 4](#) from the NFC interface and in [Table 5](#) from the I²C interface. The SRAM memory is only available and accessible when powered via VCC. Please refer to [Section 11](#) for examples of memory map from the NFC interface with SRAM mapping.

The structure of manufacturing data, static and dynamic lock bytes, capability container and user memory pages are compatible with other NTAG products.

Any memory access which starts at a valid address and extends into an invalid access region will return 00h value for the invalid region.

Bits and bytes marked as reserved for future use (RFU) SHALL NOT be changed, as it may lead to unintended tag behavior.

8.3.1 Memory map from NFC perspective

Memory access from the NFC perspective is organized in pages of 4 bytes each. If password protection is not used, complete user memory is unprotected.

8.3.2 Memory map from I²C interface

The memory access of CRN120 from the I²C interface is organized in blocks of 16 bytes each.

I²C slave address is stored in most significant 7 bits of byte 0 in block 0. However, when reading block 0, CRN120 always returns 04h for byte 0.

WARNING: When configuring Static lock bytes and Capability container, Address byte gets updated, too. Address byte consists of slave address (coded in most significant 7 bits) and least significant bit set to 0b.

REMARK: For convenience reasons it is recommended to configure Address byte (block 0, byte 0) to 04h.

Table 5. CRN120 memory organization from the I²C perspective

I ² C block address		Byte number within a block				Access conditions						
		0	1	2	3	I ² C_PROT						
		4	5	6	7							
		8	9	10	11	00b	01b	1xb				
Dec.	Hex.	12	13	14	15							
0	00h	Addr. ¹	Serial number (UID)			READ&WRITE						
		Serial number (UID)		Internal								
		Internal		Static lock bytes								
		Capability Container (CC)										
1	01h	Unprotected user memory				READ&WRITE						
...	...											
AUTH0	AUTH0	Protected user memory				READ&WRITE	READ	NAK				
...	...											
55	37h	Protected user memory				READ&WRITE	READ	NAK				
56	38h											
57	39h	Dynamic lock bytes		00h		READ&WRITE						
		RFU	RFU	RFU	AUTH0							
		ACCESS	RFU	RFU	RFU							
		PWD ²		RFU	RFU							
58	3Ah	PACK ²		RFU	RFU	see Table 9						
		PT_I2C	RFU	RFU	RFU							
59	3Bh	Configuration registers				see Table 9						
									00h	00h	00h	00h
									00h	00h	00h	00h
...	...	Invalid access - returns NAK				n.a.						
247	F7h											
248	F8h	SRAM memory (64 bytes)				READ&WRITE						
...	...											
251	FBh	Invalid access - returns NAK				n.a.						
...	...											
254	FEh	Session registers				see Table 10						
									00h	00h	00h	00h
									00h	00h	00h	00h
255	FFh	Invalid access - returns NAK				n.a.						

Table 5. CRN120 memory organization from the I²C perspective...continued

I ² C block address		Byte number within a block				Access conditions		
		0	1	2	3	I ² C_PROT		
		4	5	6	7			
		8	9	10	11			
Dec.	Hex.	12	13	14	15	00b	01b	1xb

¹ The byte 0 of block 0 is always read as 04h (UID0). Writing to block 0 updates the I²C address.

² On reading PWD and PACK, CRN120 always returns 00h for all bytes

8.3.3 EEPROM

The EEPROM is a non-volatile memory that stores the 7 byte UID, the memory lock conditions, IC configuration information and the user memory.

NXP introduced with CRN120 the possibility to split the memory in an open and a password protected area see [Section 8.3.11](#).

8.3.4 SRAM

For frequently changing data, a volatile memory of 64 bytes with unlimited endurance is built in. The 64 bytes are mapped in a similar way as done in the EEPROM, i.e., 64 bytes are seen as 16 pages of 4 bytes from NFC perspective.

The SRAM is only available when the tag is powered via the VCC pin.

The SRAM is located at the end of the memory space and it is always directly accessible by the I²C host (addresses F8h to FBh). An NFC device cannot access the SRAM memory in normal mode (i.e., outside the pass-through mode). The SRAM is only accessible by the NFC device if the SRAM is mirrored onto the EEPROM memory space.

With SRAM mirror enabled (SRAM_MIRROR_ON_OFF = 1b - see [Section 11.2](#)), the SRAM can be mirrored in the User Memory from start page 01h to 74h for access from the NFC side.

The Memory mirror must be enabled once both interfaces are ON as this feature is disabled after each POR.

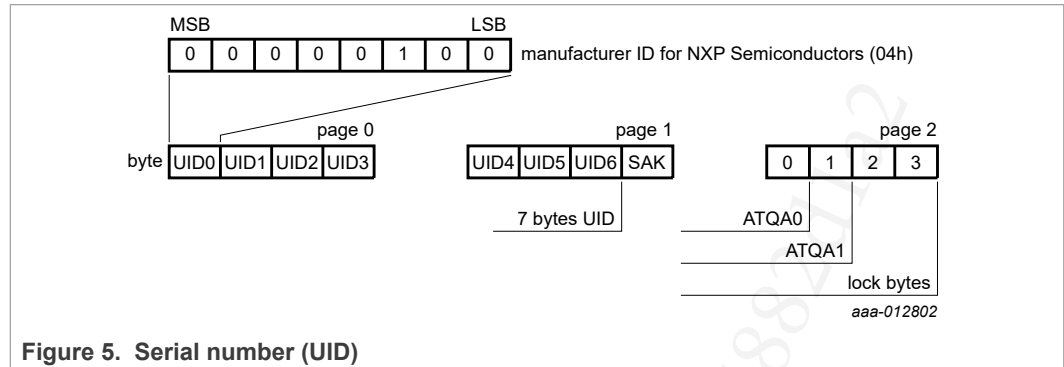
The register SRAM_MIRROR_BLOCK (see [Table 9](#)) indicates the address of the first page of the SRAM buffer. In the case where the SRAM mirror is enabled and the READ command is addressing blocks where the SRAM mirror is located, the SRAM byte values will be returned instead of the EEPROM byte values. Similarly, if the tag is not VCC powered, the SRAM mirror is disabled and reading out the bytes related to the SRAM mirror position would return the values from the EEPROM.

In the pass-through mode (PTHRU_ON_OFF = 1b - see [Table 9](#)), the SRAM is mirrored to the fixed address F0h - FFh for NFC access (see [Section 11](#)) in the first memory sector (Sector 0).

8.3.5 Serial number (UID)

The unique 7 byte serial number (UID) is programmed into the first 7 bytes of memory covering page addresses 00h and 01h - see [Figure 5](#). These bytes are programmed and write protected during production.

UID0 is fixed to the value 04h - the manufacturer ID for NXP Semiconductors in accordance with [ISO/IEC 7816-6](#).

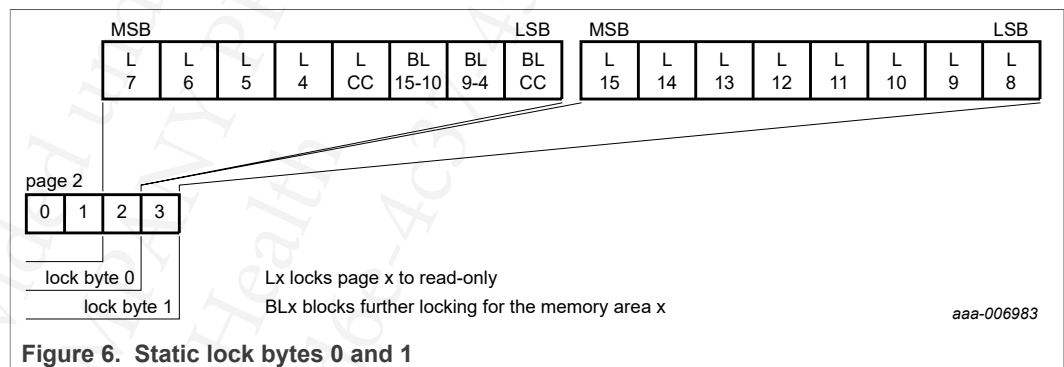


8.3.6 Static Lock Bytes

According to [NFC Forum Type 2 Tag](#) specification, the bits of byte 2 and byte 3 of page 02h (via NFC) or byte 10 and 11 address 00h (via I²C) represent the field programmable, read-only locking mechanism (see [Figure 6](#)). Each page from 03h (CC) to 0Fh can be individually locked by setting the corresponding locking bit to logic 1b to prevent further write access. After locking, the corresponding page becomes read-only memory.

This read only locking is address-based. This means, when SRAM is mirrored to these blocks, also SRAM blocks are read only from NFC perspective.

In addition, CRN120 uses the three least significant bits of lock byte 0 as the block-locking bits. Bit 2 controls pages 0Ah to 0Fh (via NFC), bit 1 controls pages 04h to 09h (via NFC) and bit 0 controls page 03h (CC). Once the block-locking bits are set, the locking configuration for the corresponding memory area is frozen, e.g. cannot be changed to read-only anymore.



For example, if BL15-10 is set to logic 1b, then bits L15 to L10 (lock byte 1, bit[7:2]) can no longer be changed. The static locking and block-locking bits are set by the bytes 2 and 3 of the WRITE command to page 02h. The contents of the lock bytes are bit-wise OR'ed and the result then becomes the new content of the lock bytes. This process is irreversible from NFC perspective. If a bit is set to logic 1b, it cannot be changed back to logic 0b. From I²C perspective, the bits can be reset to 0b by writing bytes 10 and 11 of block 00h. As I²C address is coded in byte 0 of block 0, it may be changed unintentionally.

The contents of bytes 0 and 1 of page 02h (via NFC) are unaffected by the corresponding data bytes of the WRITE command.

The default value of the static lock bytes is 0000h.

In NFC charging applications the static lock bytes should be set to 0Fh, 00h after programming the capability container.

8.3.7 Dynamic Lock Bytes

To lock the pages of CRN120 starting at page address 16 and onwards, the dynamic lock bytes are used. The dynamic lock bytes are located in Sector 0 at page E2h. The three lock bytes cover the memory area of 840 data bytes for CRN120. The granularity is 16 pages for CRN120 compared to a single page for the first 48 bytes.

CRN120 needs a Lock Control TLV as specified in [NFC Forum Type 2 Tag](#) specification to ensure NFC Forum Type 2 Tag compliancy.

When NFC Forum Type 2 Tag transition to READ ONLY state is intended, all bits marked as RFUI and dynamic lock bits related to the protected area shall be set to 0b when writing to the dynamic lock bytes.

The default value of the dynamic lock bytes is 000000h. The value of Byte 3 is always 00h.

Like for the static lock bytes, this process of modifying the dynamic lock bits is irreversible from NFC perspective and applies also for potentially mirrored SRAM. If a bit is set to logic 1b, it cannot be changed back to logic 0b. From I²C interface, these bits can be set to 0b again.

In NFC charging applications the dynamic lock bytes in block E2h should be set to 00h, 00h, 7Fh.

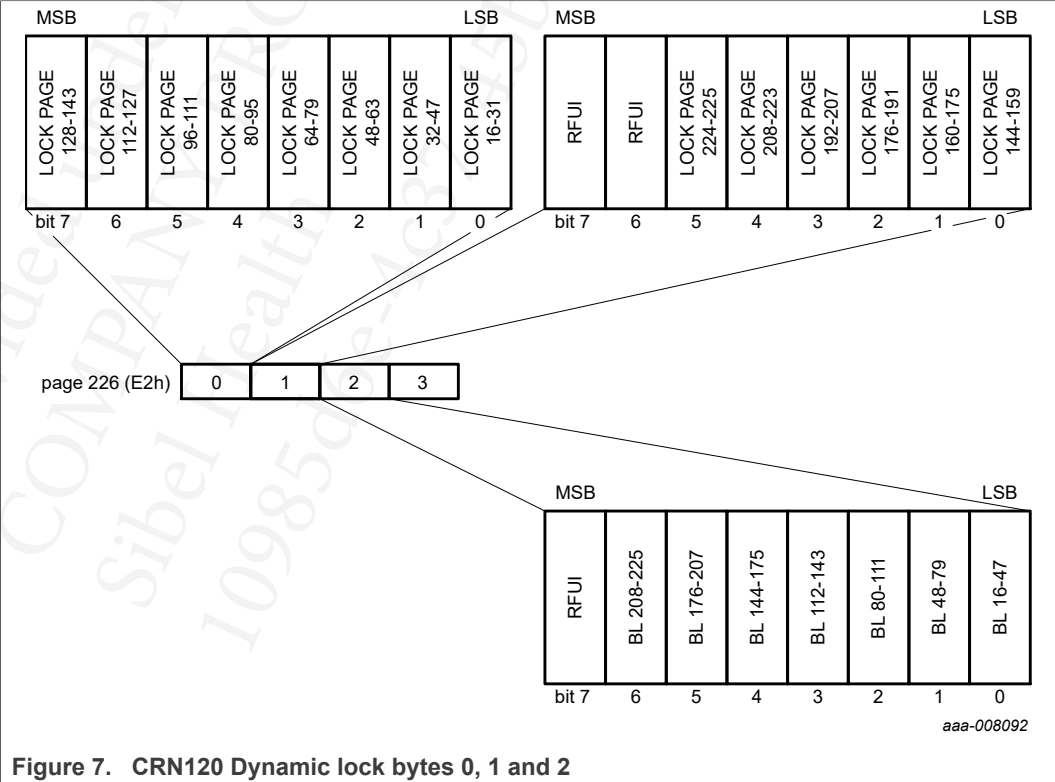


Figure 7. CRN120 Dynamic lock bytes 0, 1 and 2

8.3.8 Capability Container (CC)

According to [NFC Forum Type 2 Tag](#) specification the CC is located on page 03h. To keep full flexibility to split the memory into an open and protected area, the default value of the CC is initialized with 00000000h during the IC production.

NDEF messages can only be written with NFC Forum devices, after setting these CC bytes according to application-specific needs and NFC Forum specification by a WRITE command from the I²C or NFC interface. According to NFC Forum specification, a bit once set to 1b, an NFC Forum Device cannot set bits of the CC back to 0b. However, similar to the lock bits, setting these bits back to 0b is again possible from I²C perspective.

WARNING: As I²C address (byte 0) and static lock bytes (byte 10 and byte 11) are coded in block 00h from I²C side, the I²C address may be changed or the tag may be locked unintentionally, when changing CC.

REMARK: When reading out byte 0, CRN120 always returns 04h (UID0). Therefore, for convenience reasons it is recommended to configure I²C address byte to 04h.

NXP recommends setting the size parameter of the CC only to values that the T2T_Area ends at lock bit granularity boundaries when using only part of the memory for storing NDEF messages. Consequently T2T_Area size should be 112 + 64*N or 888 bytes with N less or equal to 13.

In [Figure 8](#) it is shown how the CC is changed when going from READ/WRITE to READ ONLY state according to NFC Forum.

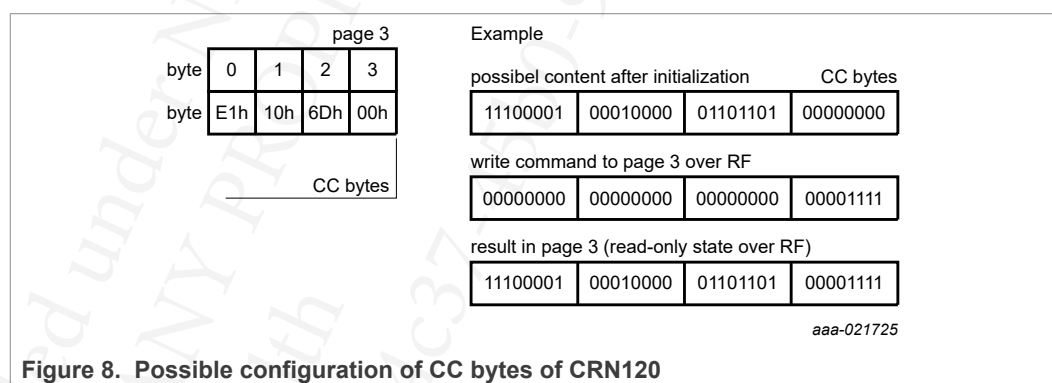


Figure 8. Possible configuration of CC bytes of CRN120

8.3.9 User Memory pages

Pages 04h to E1h of Sector 0 via the NFC interface - Block 01h to 37h, plus the first 8 bytes of block 38h via the I²C interface is the user memory area for CRN120.

8.3.10 Memory content at delivery

As described above the CC in page 03h is set to all 00h to keep the full flexibility. To allow NFC Forum NDEF message reading and writing page 03h (CC) and the following data page (NDEF TLV) of CRN120 need to be initialized by the user according to the [NFC Forum Type 2 Tag](#) specification. [Table 6](#) shows an example of NFC Forum-compliant content using the whole memory of sector 0 for NDEF messages.

Remark: The default content of the data pages from page 04h onwards is not defined at delivery.

Table 6. Minimum memory content to be in initialized state for CRN120

Page Address	Byte number within page			
	0	1	2	3
03h	E1h	10h	6Dh	00h
04h	03h	00h	FEh	00h

8.3.11 Password and Access Configuration

CRN120 can be configured to have password protected memory areas.

If this feature is used, NXP recommends changing and diversify the PWD and PACK for every single chip.

The password and access configuration area of pages E3h to E7h (Sector 0 - see [Table 7](#)) via the NFC interface or blocks 38h and 39h via the I²C interface are used to configure the password and access conditions of the CRN120. Those bit values are stored in the EEPROM. Their values can be read and written by both interfaces when applicable and when not locked by the register lock bits (see REG_LOCK in [Table 9](#)).

AUTH0 defines the starting page address of the protected area in Sector 0. NXP recommends setting AUTH0 in a way always respecting the lock bit granularity. Setting AUTH0 greater EBh, disables password protection.

The NFC_PROT bit is used to either only require a PWD_AUTH for writing data to the protected area or even protect reading data from the protected area.

If password authentication is used, even the SRAM access can be protected by setting SRAM_PROT bit to 1b.

I2C_PROT enables the possibility to limit access to the protected area from I²C perspective to read only or no access at all.

AUTLIM value can be used to limit negative PWD_AUTH attempts.

Once password protection is enabled, writing to Password and Access Configuration bytes is only possible after a successful password authentication. On reading the PWD or PACK, from NFC or I²C perspective, CRN120 always returns all 00h bytes.

A detailed description of the mechanism and how to program all the parameters is given in [Section 8.7](#).

Table 7. Password and Access Configuration Register

NFC page address (Sector 0)		I ² C block address		Byte number from NFC perspective			
Dec	Hex	Dec	Hex	0	1	2	3
224	E0h	56	38h	User Memory			
225	E1h						
226	E2h			Dynamic lock bytes			00h
227	E3h			RFU	RFU	RFU	AUTH0
228	E4h	57	39h	ACCESS	RFU	RFU	RFU
229	E5h			PWD			
230	E6h			PACK		RFU	RFU
231	E7h			PT_I2C	RFU	RFU	RFU

Table 8. Password and Access Configuration bytes

Bit	Field	Access via NFC	Access via I ² C	Default values	Description
Authentication Pointer (AUTH0)					
7-0	AUTH0	R&W	R&W	FFh	Page address of Sector 0 from which onwards the password authentication is required to access the user memory from NFC perspective, dependent on NFC_PROT bit. If AUTH0 is set to a page address greater than EBh, the password protection is effectively disabled. Password protected area starts from page AUTH0 and ends at page EBh. Password protection is excluded for Dynamic Lock Bits, session registers and mirrored SRAM pages. REMARK: From I ² C interface, you have access to all configuration pages until REG_LOCK_I2C bit is set to 1b.
Access Conditions (ACCESS)					
7	NFC_PROT	R&W	R&W	0b	Memory protection bit: 0b: write access to protected area is protected by the password 1b: read and write access to protected area is protected by the password
2-0	AUTHLIM	R&W	R&W	000b	Limitation of negative password authentication attempts. After reaching the limit, protected area is not accessible any longer. 000b: limiting of negative password authentication attempts disabled. 001b-111b: maximum number of negative password authentication attempts is 2 ^{AUTHLIM}
Password (PWD)					
31-0	PWD	R&W	R&W	FFFFFFFFh	32-bit password used for memory access protection. Reading PWD always returns 00000000h
Password Acknowledge (PACK)					
15-0	PACK	R&W	R&W	0000h	16-bit password acknowledge used during the password authentication process. Reading PACK always returns 0000h
Protection bits (PT_I2C)					

Table 8. Password and Access Configuration bytes...continued

Bit	Field	Access via NFC	Access via I ² C	Default values	Description
7-	RFU	R&W	R&W	0000b	RFU - SHALL be 0000b
2	SRAM_PROT	R&W	R&W	0b	Password protection for pass-through and mirror mode 0b: password authentication for pass-through mode disabled 1b: password authentication needed to access SRAM in pass-through mode
1-0	I2C_PROT	R&W	R&W	00b	Access to protected area from I ² C perspective 00b: Entire user memory accessible from I ² C 01b: read and write access to unprotected user area, read only access to protected area 1Xb: read and write access to unprotected area, no access to protected area. REMARK: Independent from these bits I ² C always has R&W access to: • Session registers • SRAM • Configuration pages including PWD Configuration area, but dependent on REG_LOCK_I2C bit

8.3.12 CRN120 configuration and session registers

CRN120 behavior can be configured and read in two separate locations depending if the configurations shall be effective within the communication session (use session registers) or by default after Power-On Reset (POR) (use configuration registers).

The configuration registers of pages E8h to E9h (Sector 0 - see [Table 9](#)) via the NFC interface or block 3Ah via the I²C interface are used to configure the default behavior of the CRN120. Those bit values are stored in the EEPROM and represent the default settings to be effective after POR. Their values can be read and written by both interfaces when applicable and when not locked by the register lock bits (see REG_LOCK in [Table 11](#)).

For NFC charging applications, it is recommended to set REG_LOCK byte to 03h as last step of personalization.

Table 9. Configuration register CRN120

NFC address (Sector 0)		I ² C Address		Byte number from NFC perspective			
Dec	Hex	Dec	Hex	0	1	2	3
232	E8h	58	3Ah	NC_REG	LAST_NDEF_BLOCK	SRAM_MIRROR_BLOCK	WDT_LS
233	E9h			WDT_MS	I2C_CLOCK_STR	REG_LOCK	RFU

The session register on pages ECh to EDh (Sector 0) via the NFC interface or block FEh via I²C, see [Table 10](#), are used to configure or monitor the values of the current communication session. Those bits are read only via the NFC interface but may be read and written via the I²C interface.

For backward compatibility reasons, the session registers are mirrored to Sector 3 (page F8h and F9h via the NFC interface).

Table 10. Session registers CRN120

NFC address (Sector 0)		I ² C Address		Byte number			
Dec	Hex	Dec	Hex	0	1	2	3
236	ECh	254	FEh	NC_REG	LAST_NDEF_BLOCK	SRAM_MIRROR_BLOCK	WDT_LS
237	EDh			WDT_MS	I2C_CLOCK_STR	NS_REG	RFU

Both, the session and the configuration registers have the same configuration options and parameters except the REG_LOCK bits, which are only available in the configuration register and the NS_REG bits which are only available in the session register. After POR, the content of the configuration register is loaded into the session register.

The values of both registers can be changed during a communication session. If the desired effect should be visible immediately, but only for the current communication session, the session registers must be used. After POR, the session registers values will again contain the configuration register values as before.

To change the default behavior, changes to the configuration register are needed, but the related effect will only be visible after the next POR.

To make the effect immediately and after next POR visible, changes to configuration and session registers are needed.

All registers and configuration default values, access conditions and descriptions are defined in [Table 11](#) and [Table 12](#).

Reading and writing the session registers via I²C can only be done via the READ and WRITE registers operation - see [Section 9.8](#).

Table 11. Configuration bytes

Bit	Field	Access via NFC	Access via I ² C	Default values	Description
Configuration register: NC_REG					
7	NFCS_I2C_RST_ON_OFF	R&W	R&W	0b	Enables the NFC disable feature and enables soft reset through I ² C repeated start - see Section 9.3
6	PTHRU_ON_OFF	R&W	R&W	0b	1b: pass-through mode using SRAM enabled and SRAM mapped to end of Sector 0. 0b: pass-through mode disabled
5-4	ED_OFF	R&W	R&W	00b	defines the event upon which the signal output on the ED pin is released 00b: if the field is switched off 01b: if the field is switched off or the tag is set to the HALT state 10b: if the field is switched off or the last page of the NDEF message has been read (defined in LAST_NDEF_BLOCK) 11b: (if ED_ON = 11b) if the field is switched off or if last data is read by I ² C (in pass-through mode NFC ---> I ² C) or last data is written by I ² C (in pass-through mode I ² C----> NFC) 11b: (if ED_ON = 00b or 01b or 10b) if the field is switched off See Section 8.4 for more details
3-2	ED_ON	R&W	R&W	00b	defines the event upon which the signal output on the ED pin is pulled low 00b: RFU 01b: by first valid start of communication (SoC) 10b: by selection of the tag 11b: (in pass-through mode NFC-->I ² C) if the data is ready to be read from the I ² C interface 11b: (in pass-through mode I ² C--> NFC) if the data is read by the NFC interface See Section 8.4 for more details
1	SRAM_MIRROR_ON_OFF	R&W	R&W	0b	1b: SRAM mirror enabled and mirrored SRAM starts at page SRAM_MIRROR_BLOCK 0b: SRAM mirror disabled
0	TRANSFER_DIR	R&W	R&W	1b	defines the data flow direction when pass-through mode is enabled 0b: from I ² C to NFC interface 1b: from NFC to I ² C interface In case the pass-through mode is NOT enabled, this bit should be set to 1b, otherwise there is no WRITE access from the NFC perspective
Configuration register: LAST_NDEF_BLOCK					

Table 11. Configuration bytes...continued

Bit	Field	Access via NFC	Access via I ² C	Default values	Description
7-0	LAST_NDEF_BLOCK	R&W	R&W	00h	I ² C block address of I ² C block, which contains last byte(s) of stored NDEF message. An NFC read of the last page of this I ² C block sets the register NDEF_DATA_READ to 1b and triggers field detection pin if ED_OFF is set to 10b. Valid range starts from 01h (NFC page 04h) up to 37h (NFC page DCh) CRN120
Configuration register: SRAM_MIRROR_BLOCK					
7-0	SRAM_MIRROR_BLOCK	R&W	R&W	F8h	I ² C block address of SRAM when mirrored into the User memory. Valid range starts from 01h (NFC page 04h) up to 34h (NFC page D0h) CRN120
Configuration register: WDT_LS					
7-0	WDT_LS	R&W	R&W	48h	Least Significant byte of watchdog time control register
Configuration register: WDT_MS					
7-0	WDT_MS	R&W	R&W	08h	Most Significant byte of watchdog time control register. When writing WDT_MS byte, the content of WDT_MS and WDT_LS gets active for the watchdog timer.
Configuration register: I2C_CLOCK_STR					
7-1	RFU	R&W	R&W	0000000b	RFU - all 7 bits SHALL be 0b
0	I2C_CLOCK_STR	R&W	R&W	1b	Enables (1b) or disable (0b) the I ² C clock stretching
Configuration register: REG_LOCK					
7-2	RFU	R&W	R&W	000000b	RFU - all 6 bits SHALL be 0b
1	REG_LOCK_I2C ¹	R&W	R&W	0b	I ² C Configuration Lock Bit 0b: Configuration bytes may be changed via I ² C 1b: Configuration bytes cannot be changed via I ² C Once set to 1b, cannot be reset to 0b anymore.
0	REG_LOCK_NFC ¹	R&W	R&W	0b	NFC Configuration Lock Bit 0b: Configuration bytes may be changed via NFC 1b... Configuration bytes cannot be changed via NFC Once set to 1b, cannot be reset to 0b anymore.

¹ Setting both bits REG_LOCK_I2C and REG_LOCK_NFC to 1b, permanently locks write access to register default values (as no write is allowed anymore). As long as one bit is still 0b, the corresponding interface can still access and change the register lock bytes.

Table 12. Session register bytes

Bit	Field	Access via NFC	Access via I ² C	Default values	Description
Session register: NC_REG					
7	NFCS_I2C_RST_ON_OFF	READ	R&W	-	see configuration bytes description
6	PTHRU_ON_OFF	READ	R&W	-	see configuration bytes description, the bit is cleared automatically, when one of the interfaces is OFF
5-4	ED_OFF	READ	R&W	-	see configuration bytes description
3-2	ED_ON	READ	R&W		
1	SRAM_MIRROR_ON_OFF	READ	R&W	-	see configuration bytes description, the bit is cleared automatically, when there is no Vcc power.
0	TRANSFER_DIR	READ	R&W		see configuration bytes description
Session register: LAST_NDEF_BLOCK					
7-0	LAST_NDEF_BLOCK	READ	R&W	-	see configuration bytes description
Session register: SRAM_MIRROR_BLOCK					
7-0	SRAM_MIRROR_BLOCK	READ	R&W	-	see configuration bytes description
Session register: WDT_LS					
7-0	WDT_LS	READ	R&W	-	see configuration bytes description
Session register: WDT_MS					
7-0	WDT_MS	READ	R&W	-	see configuration bytes description
Session register: I2C_CLOCK_STR					
7-2	RFU	READ	READ	-	RFU, all 6 bits locked to 0b
1	NEG_AUTH_REACHED	READ	READ	0b	Status bit to show the number of negative PWD_AUTH attempts reached 0b: PWD_AUTH still possible 1b: PWD_AUTH locked
0	I2C_CLOCK_STR	READ	READ	-	See configuration bytes description
Session register: NS_REG					
7	NDEF_DATA_READ	READ	READ	0b	1b: all data bytes read from the address specified in LAST_NDEF_BLOCK. Bit is reset to 0b when read
6	I2C_LOCKED	READ	R&W	0b	1b: Memory access is locked to the I ² C interface
5	RF_LOCKED	READ	READ	0b	1b: Memory access is locked to the NFC interface
4	SRAM_I2C_READY	READ	READ	0b	1b: data is ready in SRAM buffer to be read by I2C
3	SRAM_RF_READY	READ	READ	0b	1b: data is ready in SRAM buffer to be read by NFC
2	EEPROM_WR_ERR	READ	R&W	0b	1b: HV voltage error during EEPROM write or erase cycle Needs to be written back via I ² C to 0b to be cleared

Table 12. Session register bytes...continued

Bit	Field	Access via NFC	Access via I ² C	Default values	Description
1	EEPROM_WR_BUSY	READ	READ	0b	1b: EEPROM write cycle in progress - access to EEPROM disabled 0b: EEPROM access possible
0	RF_FIELD_PRESENT	READ	READ	0b	1b: NFC field is detected

8.4 Configurable Event Detection Pin

The event detection pin based on open-drain implementation provides the capability to trigger an external device (e.g. μ Controller) or switch on the connected circuitry by an external power management unit depending on activities on the NFC interface.

As the event detection pin functionality is operated via NFC field power, V_{CC} supply for the tag itself is not required.

The conditions for pulling the event detection signal to low, ED_ON can be:

- The detection of a valid command (Start of Communication)
- The selection of the IC

The conditions for releasing the event detection signal defined with ED_OFF can be:

- The absence of the NFC field
- The detection of the HALT state
- The NFC interface has read the last part of the NDEF message defined with LAST_NDEF_BLOCK

All the various combinations of configurations are described in [Table 11](#) and illustrated in the Figures below for all various combinations of the event detection signal configuration. The timing diagrams are not in scale and all given timing values are typical values.

The event detection pin can be used also as a handshake mechanism in the pass-through mode to signal to the external μ Controller if

- New data is written to SRAM on the NFC interface
- Data written to SRAM from the μ Controller is read via the NFC interface.

See [Section 11](#) for more information on this handshake mechanism.

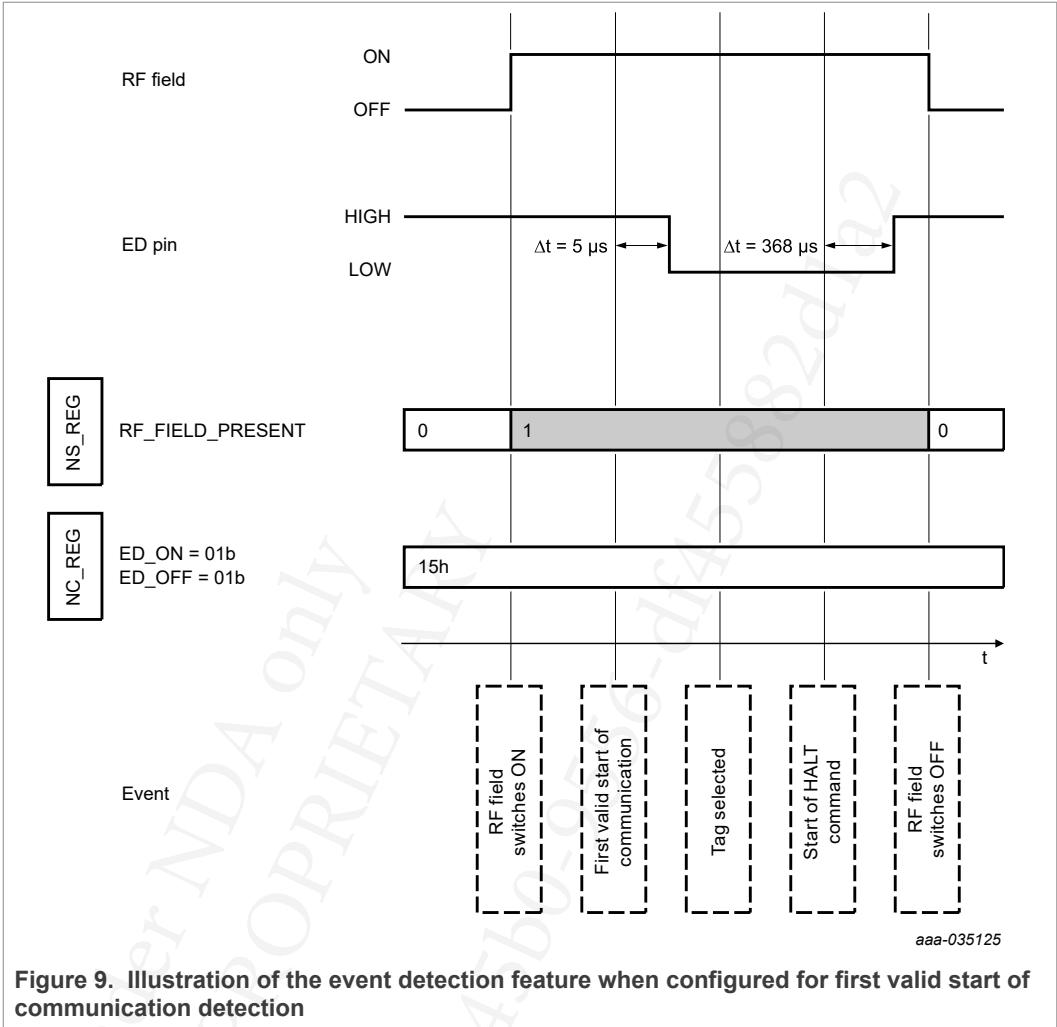


Figure 9. Illustration of the event detection feature when configured for first valid start of communication detection

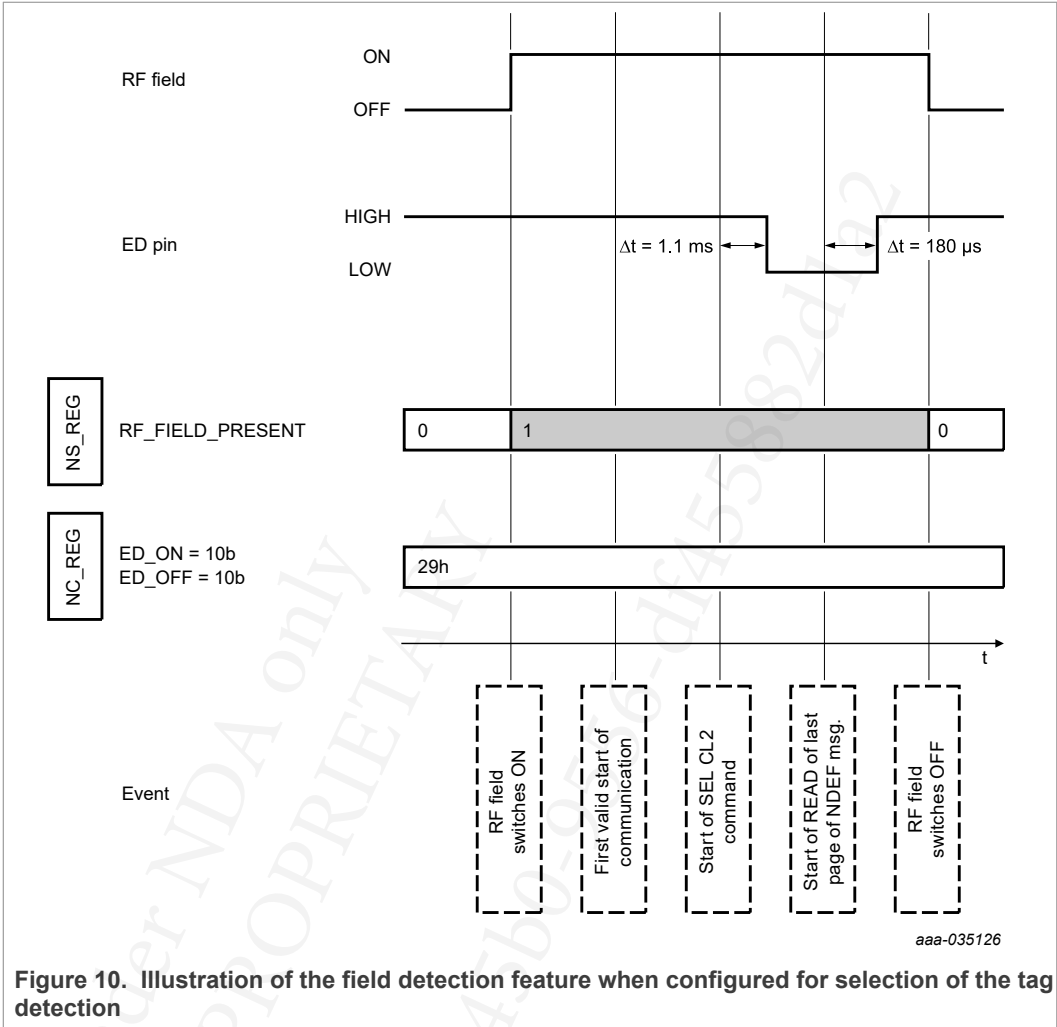


Figure 10. Illustration of the field detection feature when configured for selection of the tag detection

8.5 Field Detection Pin

FD pin can be used as a trigger for the host system to indicate the presence of the NFC field.

The voltage at the FD pin is defined in the characteristics table.

A decoupling cap of 4.7 nF and a pull-down resistor of 47 kΩ shall be used.

The maximum delay from NFC field ON to FD pin HIGH is typically 30 μs.

The maximum delay from NFC field OFF to FD pin LOW is typically 200 μs for below example circuit without load.

For a 1.8 V I/O a serial voltage divider resistor needs to be added and adjusted in relation to the MCU-integrated pull-down.

8.6 Watchdog timer

In order to allow the I²C interface to perform all necessary commands (READ, WRITE, ..), the memory access remains locked to the I²C interface until the register I2C_LOCKED is cleared by the host - see [Table 11](#).

However, to avoid that the memory stays 'locked' to the I²C for a long period of time, it is possible to program a watchdog timer to unlock the I²C host from the tag, so that the NFC device can access the tag after a period of time of inactivity. The host itself will not be notified of this event directly, but the NS_REG register is updated accordingly (the register bit I2C_LOCKED will be cleared - see [Table 11](#)).

The default value is set to 20 ms (848h), but the watchdog timer can be freely set from 0001h (9.43 μ s) up to FFFFh (617.995 ms). The timer starts ticking when the communication between the CRN120 and the I²C interface starts. In case the communication with the I²C is still going on after the watchdog timer expires, the communication will continue until the communication has completed. Then the status register I2C_LOCKED will be immediately cleared.

In the case where the communication with the I²C interface has completed before the end of the timer and the status register I2C_LOCKED was not cleared by the host, it will be cleared at the end of the watchdog timer.

The watchdog timer is only effective if the VCC pin is powered and will be reset and stopped if the CRN120 is not VCC powered or if the register status I2C_LOCKED is set to 0 and RF_LOCKED is set to 1b.

8.7 Password authentication

The memory write or read/write access to a configurable part of the memory can be constrained to a positive password authentication. The 32-bit secret password (PWD) and the 16-bit password acknowledge (PACK) response shall be typically programmed into the configuration pages at the tag personalization stage.

The AUTHLIM parameter specified in [Table 8](#) can be used to limit the negative authentication attempts.

In the initial state of CRN120, password protection is disabled by an AUTH0 value of FFh. PWD and PACK are freely writable in this state. Access to the configuration pages and any part of the user memory can be restricted by setting AUTH0 to a page address within the available memory space. This page address is the first one protected.

For a comprehensive description of all protection mechanism, refer to the [application note](#).

Remark: The password protection method provided in CRN120 has to be intended as an easy and convenient way to prevent unauthorized memory accesses. If a higher level of protection is required, cryptographic methods can be implemented at application layer to increase overall system security.

8.7.1 Programming of PWD and PACK

The 32-bit PWD and the 16-bit PACK need to be programmed into the configuration pages, see [Table 7](#). The password as well as the password acknowledge are written LSByte first. This byte order is the same as the byte order used during the PWD_AUTH command and its response.

The PWD and PACK bytes can never be read out of the memory. Instead of transmitting the real value on any valid read command from both - NFC and I²C - interface, only 00h bytes are replied.

If the password authentication is disabled, PWD and PACK can be written at any time.

If the password authentication is enabled, PWD and PACK can be written after a successful PWD_AUTH command only.

Remark: To improve the overall system security, it is advisable to diversify the password and the password acknowledge using a die individual parameter of the IC, which can be the 7 byte UID available on CRN120.

8.7.2 Limiting negative verification attempts

To prevent brute-force attacks on the password, the maximum allowed number of negative password authentication attempts can be set using AUTHLIM. This mechanism is disabled by setting AUTHLIM to a value of 000b, which is also the initial state of CRN120.

If AUTHLIM is not equal to 000b, each negative authentication verification is internally counted. As soon as this internal counter reaches the number 2^{AUTHLIM} , any further negative password authentication leads to a permanent locking of the protected part of the memory for the specified access modes. Independently, whether the provided password is correct or not, each subsequent PWD_AUTH fails.

Any successful password verification, before reaching the limit of negative password verification attempts, resets the internal counter to zero.

8.7.3 Protection of configuration segments

The configuration pages can be protected by the password authentication as well. The protection level is defined with the NFC_PROT bit.

The protection is enabled by setting the AUTH0 byte (see [Table 7](#)) to a value that is within the addressable memory space.

8.8 NFC charging receiver block diagram

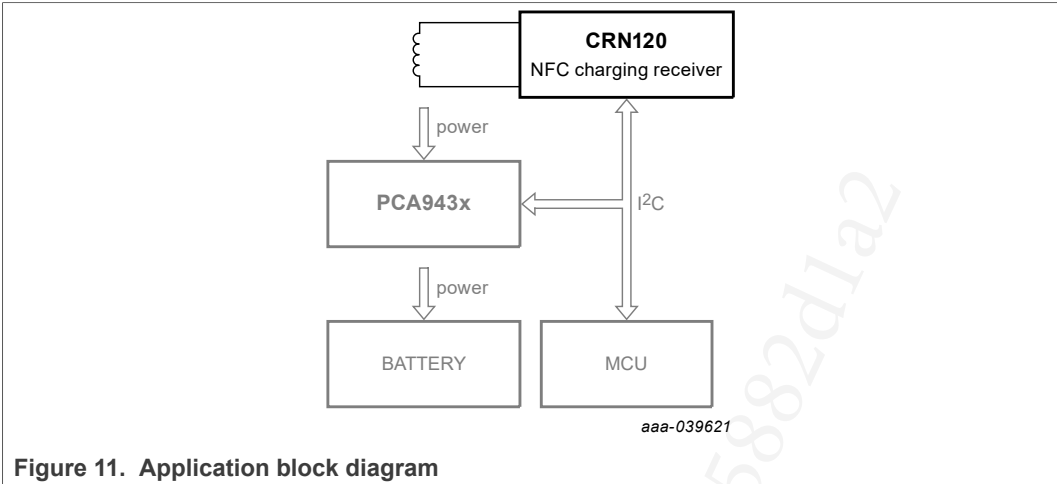
In the figure below, a high-level block diagram of an NFC charging receiver using CRN120 is illustrated.

A highly efficient Schottky rectifier is used to power the complete system out of the NFC field. [PCA9430](#) or [PCA9431](#) should be used to charge the battery directly or with the help of a power management IC. The NFC Forum defined communication link is established by using CRN120.

Reference schematics and further details can be found in NXP's NFC charging related application notes - see [AN12639](#) and [AN12641](#).

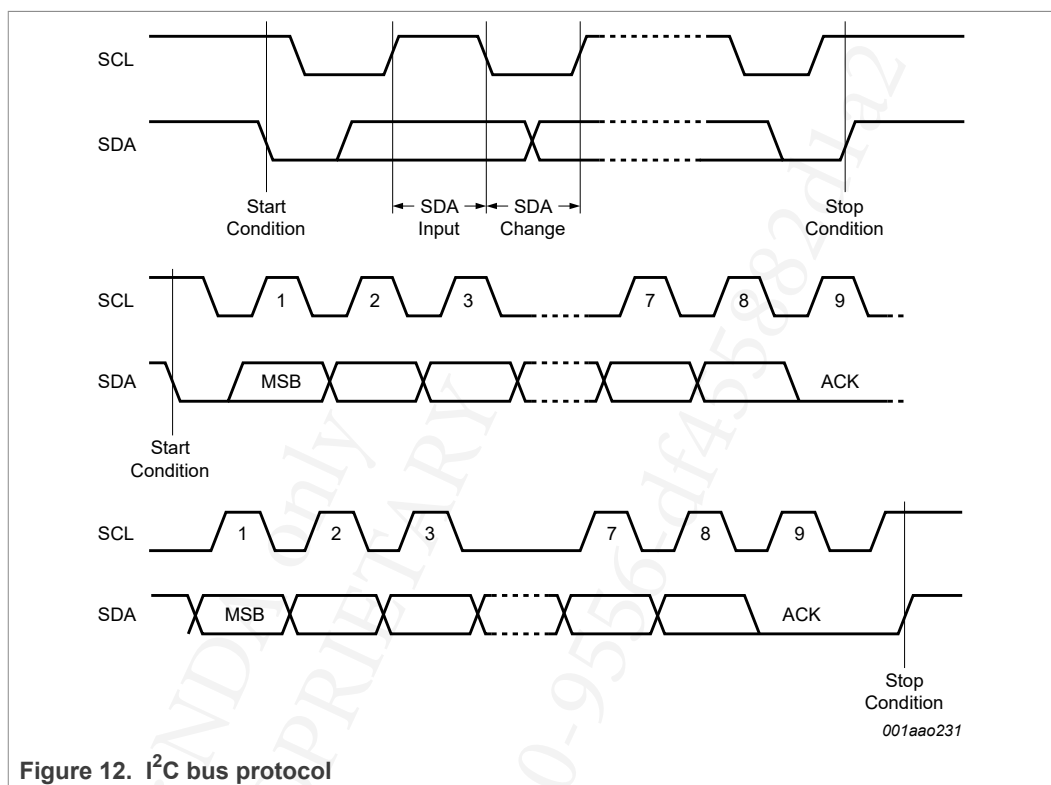
The NFC charging development kit is described in [UM11335](#).

NFC charging software and tools as well as libraries are described in [UM11336](#) and [SW5907](#).



9 I²C commands

For details about I²C interface, refer to the [NXP user manual](#).



The CRN120 supports the I²C protocol. This protocol is summarized in [Figure 12](#). Any device that sends data onto the bus is defined as a transmitter, and any device that reads the data from the bus is defined as a receiver. The device that controls the data transfer is known as the "bus master", and the other as the "slave" device. A data transfer can only be initiated by the bus master, which will also provide the serial clock for synchronization. The CRN120 is always a slave in all communications.

9.1 Start condition

Start is identified by a falling edge of Serial Data (SDA), while Serial Clock (SCL) is stable in the high state. A Start condition must precede any data transfer command. The CRN120 continuously monitors SDA (except during a Write cycle) and SCL for a Start condition, and will not respond unless one is given.

9.2 Stop condition

Stop is identified by a rising edge of SDA while SCL is stable and driven high. A Stop condition terminates communication between the CRN120 and the bus master. A Stop condition at the end of a Write command triggers the internal Write cycle.

WARNING: Host shall respect EEPROM programming time (~4 ms) after this Stop condition in any case. If host sends next command too early, the memory may be corrupted as ongoing EEPROM write cycle might get terminated.

9.3 I²C soft reset and NFC disable feature

With the bit `NFCS_I2C_RST_ON_OFF` (see [Table 11](#)) CRN120 enables two features: a soft reset of the I²C subsystem, and NFC disable, in which the NFC demodulator is disabled.

The I²C soft reset feature interprets an I²C repeated start (no I²C stop in between) as a command to execute a soft reset of the I²C subsystem. This is useful when heavy bus interference can cause the I²C interface to get stuck. A drawback of this feature is that every start symbol then has to be terminated with a Stop, slowing down communication. If a Stop is forgotten, the I²C interface is cleared and previous communication, if any, is lost. Consequently when this feature is used, stop conditions after MEMA for READ/WRITE (see [Figure 13](#)) and after REGA for READ/WRITE registers (see [Figure 14](#)) shall be send.

The NFC disable feature disables the demodulator. When feature is set, no NFC commands are received, and no replies are issued to commands that were not fully received when NFC disable was set. This feature allows the tag to "disappear" even if it still is in the reader field. CRN120 will remain in the NFC state it was in when NFC disable was enabled, until NFC disable is cleared.

The combination of these two features in a single bit means that I²C soft reset is only active during NFC disable.

9.4 Acknowledge bit (ACK)

The acknowledge bit is used to indicate a successful byte transfer. The bus transmitter, whether it is the bus master or slave device, releases Serial Data (SDA) after sending 8 bits of data. During the ninth clock pulse period, the receiver pulls Serial Data (SDA) low to acknowledge the receipt of the 9th data bits.

9.5 Data input

During data input, the CRN120 samples SDA on the rising edge of SCL. For correct device operation, SDA must be stable during the rising edge of SCL, and the SDA signal must change only when SCL is driven low.

9.6 Addressing

To start communication between a bus master and the CRN120 slave device, the bus master must initiate a Start condition (see [Section 9.1](#)). Following this initiation, the bus master sends the 7-bit device address, called Slave Address (SA) in following figures.

The 8th bit is the Read/Write bit (R/W). This bit is set to 1b for Read and 0b for Write operations.

Default device address of 55h results in AAh default I²C write address and ABh default I²C read address.

As from I²C perspective I²C address can be configured via byte 0 of block 0. Reading this block gives 04h, as it is returning UID0 (see [Section 8.3.2](#)). Therefore it is recommended to us 04h as I²C write address (02h device address).

NOTE: Byte 0 of block 0 is used to configure the device address. The 7-bit device address needs to be programmed in the 7 most significant bits of this byte. Least

significant bit needs to be set to 0b when programming the device address. E.g. to keep default device address of 55h, byte 0 of block 0 needs to be set to AAh.

If a match occurs on the device address, the CRN120 gives an acknowledgment on SDA during the 9th bit time. If the CRN120 address does not match, it deselects itself from the bus and clears the register I2C_LOCKED (see [Table 12](#)).

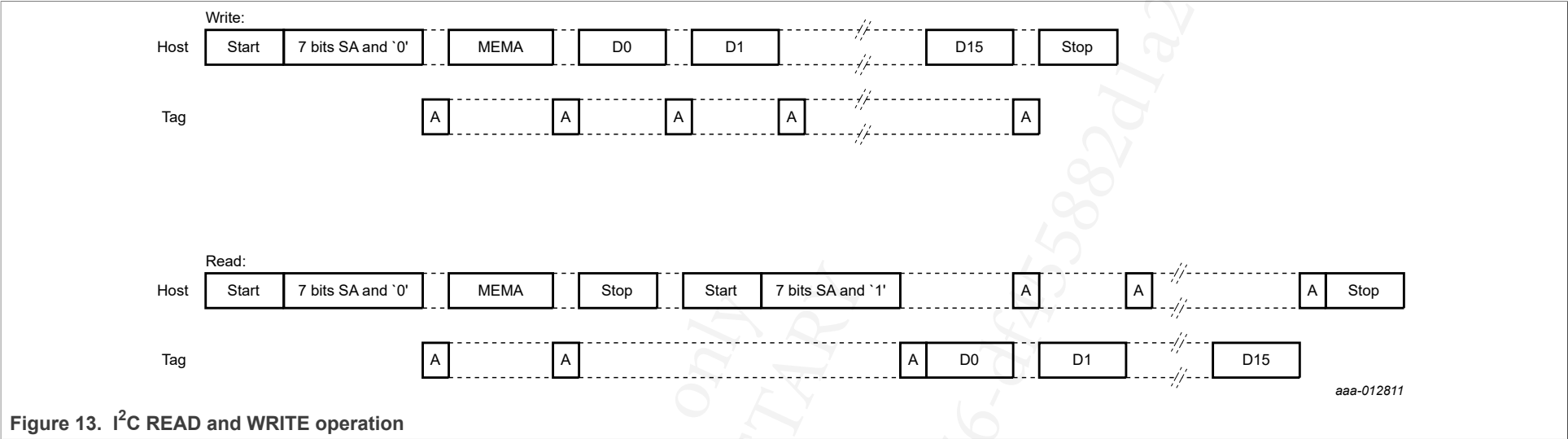
Table 13. Default CRN120 address from I²C

	Device / Slave Address (SA)							R/W
	b7	b6	b5	b4	b3	b2	b1	
Value ^[1]	1	0	1	0	1	0	1	1/0

[1] Initial values can be changed from I²C perspective

The I²C address of the CRN120 (byte 0 - block 0h) can only be modified by the I²C interface. Both interfaces cannot read the device address and a READ command from the NFC or I²C interface to this byte will return 04h (UID 0 - manufacturer ID for NXP Semiconductors - see [Figure 5](#)).

9.7 READ and WRITE Operation



The READ and WRITE operation always handle 16 bytes to be read or written (one block - see [Table 5](#))

For the READ operation (see [Figure 13](#)), following a Start condition, the bus master/host sends the CRN120 slave address code (SA - 7 bits) with the Read/Write bit (R/W) set to 0b. The CRN120 acknowledges this (A), and waits for one address byte (MEMA), which should correspond to the address of the block of memory (SRAM or EEPROM) that is intended to be read. The CRN120 responds to a valid address byte with an acknowledge (A). A Stop condition can be then issued. Then the host again issues a start condition followed by the CRN120 slave address with the Read/Write bit set to 1b. When I2C_CLOCK_STR is set to 0b, a pause of at least 50 μ s shall be kept before this start condition. The CRN120 acknowledges this (A) and sends the first byte of data read (D0). The bus master/host acknowledges it (A) and the CRN120 will subsequently transmit the following 15 bytes of memory read with an acknowledge from the host after every byte. After the last byte of memory data has been transmitted by the CRN120, the bus master/host will acknowledge it and issue a Stop condition.

WARNING: READ sequence shall be atomic. Complete sequence of above figure needs to be executed, otherwise that tag may go to undefined state and stretches the clock infinitely.

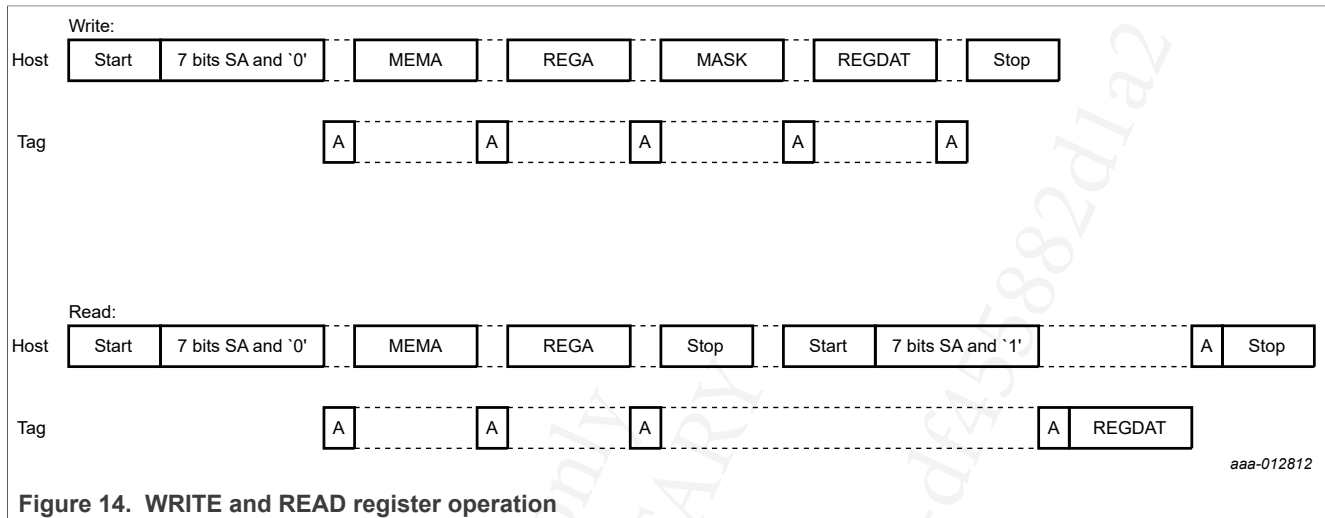
For the WRITE operation (see [Figure 13](#)), following a Start condition, the bus master/host sends the CRN120 slave address code (SA - 7 bits) with the Read/Write bit (R/W) set to 0b. The CRN120 acknowledges this (A), and waits for one address byte (MEMA), which should correspond to the address of the block of memory (SRAM or EEPROM) that is intended to be written. The CRN120 responds to a valid address byte with an acknowledge (A) and, in the case of a WRITE operation, the bus master/host starts transmitting every 16 bytes (D0...D15) that shall be written at the specified address with an acknowledge of the CRN120 after each byte (A). After the last byte acknowledge from the CRN120, the bus master/host issues a Stop condition.

WARNING: Host shall respect EEPROM programming time (~4 ms) after this Stop condition in any case. If host sends next command too early, the memory may be corrupted as ongoing EEPROM write cycle will get terminated.

The memory address accessible via the READ and WRITE operations can only correspond to the EEPROM or SRAM (respectively 00h to 3Ah or F8h to FBh for CRN120).

9.8 WRITE and READ register operation

In order to modify or read the session register bytes (see [Table 10](#)), CRN120 requires the WRITE and READ register operation (see [Figure 14](#)).



For the READ register operation, following a Start condition the bus master/host sends the CRN120 slave address code (SA - 7 bits) with the Read/Write bit (R/W) set to 0b. The CRN120 acknowledges this (A), and waits for one address byte (MEMA) which corresponds to the address of the block of memory with the session register bytes (FEh). The CRN120 responds to the address byte with an acknowledge (A). Then the bus master/host issues a register address (REGA), which corresponds to the address of the targeted byte inside the block FEh (00h, 01h...to 07h) and then waits for the Stop condition.

Then the bus master/host again issues a start condition followed by the CRN120 slave address with the Read/Write bit set to 1b. The CRN120 acknowledges this (A), and sends the selected byte of session register data (REGDAT) within the block FEh. The bus master/host will acknowledge it and issue a Stop condition.

WARNING: READ sequence shall be atomic. Complete sequence of above figure needs to be executed, otherwise that tag may go to undefined state and stretches the clock infinitely.

For the WRITE register operation, following a Start condition, the bus master/host sends the CRN120 slave address code (SA - 7 bits) with the Read/Write bit (R/W) set to 0b. The CRN120 acknowledges this (A), and waits for one address byte (MEMA), which corresponds to the address of the block of memory within the session register bytes (FEh). After the CRN120 acknowledge (A), the bus master/host issues a register address (REGA), which corresponds to the address of the targeted byte inside the block FEh (00h, 01h...to 07h). After acknowledgment (A) by CRN120, the bus master/host issues a MASK byte that defines exactly which bits shall be modified by a 1b bit value at the corresponding bit position. Following the CRN120 acknowledge (A), the new register data (one byte - REGDAT) to be written is transmitted by the bus master/host. The CRN120 acknowledges it (A), and the bus master/host issues a stop condition.

10 NFC Command

CRN120 activation follows the [NFC Forum Activity](#) specification. After CRN120 has been selected, it can either be deactivated using the NFC Forum SLP_REQ command, or NTAG commands (e.g. PWD_AUTH, SECTOR_SELECT, READ or WRITE) can be performed. For more details about the card activation, refer to [NFC Forum Activity](#) specification.

10.1 CRN120 command overview

All available commands for CRN120 are shown in [Table 14](#).

Table 14. Command overview

Command ^[1]	NFC Forum	Command code (hexadecimal)
Request	SENS_REQ	26h (7 bit)
Wake-up	ALL_REQ	52h (7 bit)
Anticollision CL1	SDD_REQ CL1	93h 20h
Select CL1	SEL_REQ CL1	93h 70h
Anticollision CL2	SDD_REQ CL2	95h 20h
Select CL2	SEL_REQ CL2	95h 70h
Halt	SLP_REQ	50h 00h
GET_VERSION	-	60h
READ	READ	30h
FAST_READ	-	3Ah
WRITE	WRITE	A2h
FAST_WRITE	-	A6h
PWD_AUTH	-	1Bh

[1] Unless otherwise specified, all commands use the coding and framing as described in [Digital Protocol](#) Specification

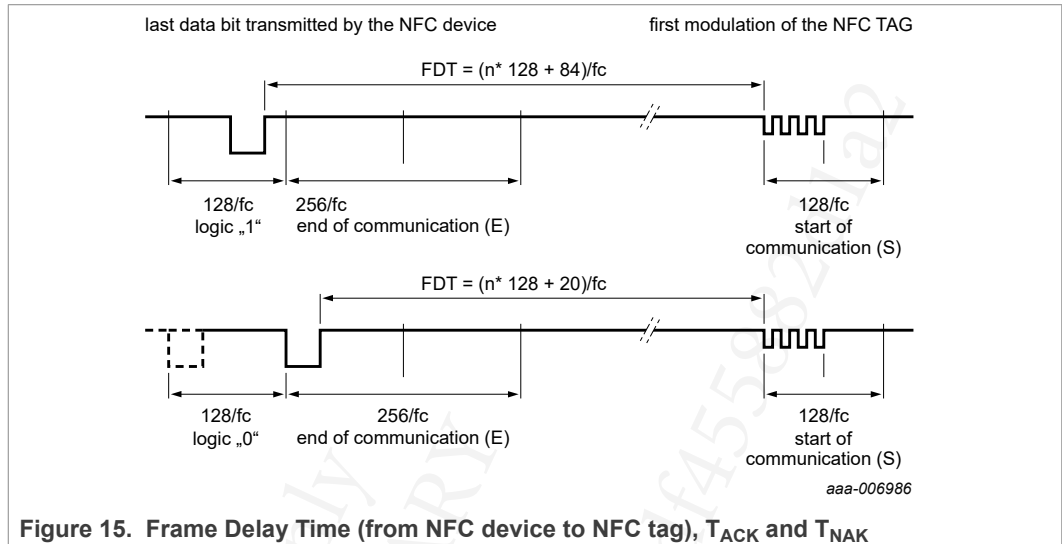
10.2 Timing

The command and response timing shown in this document are not to scale and values are rounded to 1 μ s.

All given command and response times refer to the data frames, including start of communication and end of communication. They do not include the encoding (like the Miller pulses). An NFC device data frame contains the start of communication (1 "start bit") and the end of communication (one logic 0 + 1-bit length of unmodulated carrier). An NFC tag data frame contains the start of communication (1 "start bit") and the end of communication (1-bit length of no subcarrier).

The minimum and maximum command response time is specified according to the standard. The minimum frame delay time from NFC tag to NFC device is 86.43 μ s. The maximum command response time is specified as a timeout value. Depending on the command, the T_{ACK} value specified for command responses defines the NFC device to NFC tag frame delay time. It does it for either the 4-bit ACK value specified or for a data frame.

All timing can be measured according to the NFC Forum frame specification as shown for the Frame Delay Time in [Figure 15](#). For more details, refer to [NFC Forum](#) specification.



Remark: Due to the coding of commands, the measured timings usually exclude (a part of) the end of communication. Consider this factor when comparing the specified with the measured times.

10.3 NTAG ACK and NAK

CRN120 uses a 4-bit ACK / NAK as shown in [Table 15](#).

Table 15. ACK and NAK values

Code (4 bit)	ACK/NAK
Ah	Acknowledge (ACK)
0h	NAK for invalid argument (i.e. invalid page address or wrong password)
1h	NAK for parity or CRC error
3h	NAK for Arbiter locked to I ² C
4h	Number of negative PWD_AUTH commands limit reached
7h	NAK for EEPROM write error

10.4 SENS_RES and SEL_RES responses

CRN120 replies to a SENS_REQ or ALL_REQ command with the SENS_RES value shown below. It replies to a SEL_REQ CL2 command with the SEL_RES value shown below. The 2 byte SENS_RES value is transmitted with the least significant byte first (44h).

Table 16. SENS_RES response of the CRN120

Sales type	Hex value	Bit number															
		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CRN120	00 44h	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0

Table 17. SEL_RES response of the CRN120

Sales type	Hex value	Bit number							
		7	6	5	4	3	2	1	0
CRN120	00h	0	0	0	0	0	0	0	0

10.5 GET_VERSION

The GET_VERSION command is used to retrieve information about the NTAG family, the product version, storage size and other product data required to identify the specific CRN120.

This command is also available on other NTAG products to have a common way of identifying products across platforms and evolution steps.

The GET_VERSION command has no arguments and returns the version information for the specific CRN120 type. The command structure is shown in [Figure 16](#) and [Table 18](#).

[Table 19](#) shows the required timing.

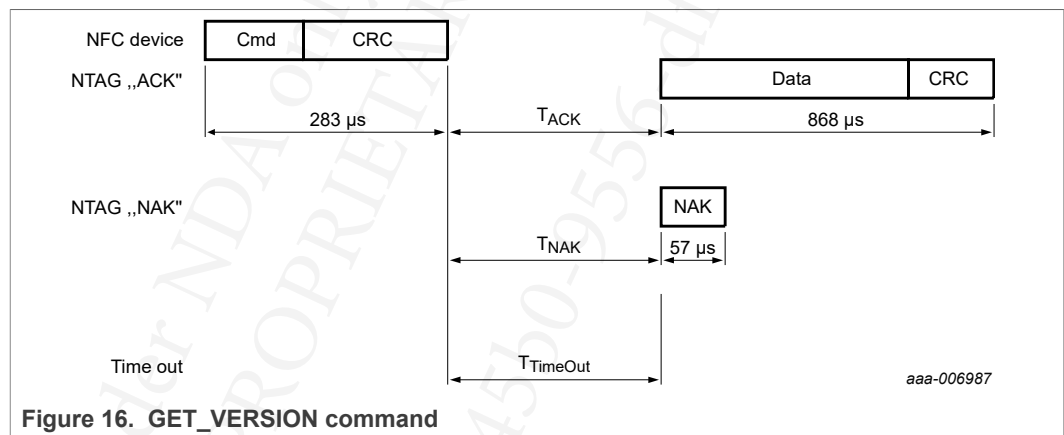


Figure 16. GET_VERSION command

Table 18. GET_VERSION command

Name	Code	Description	Length
Cmd	60h	Get product version	1 byte
CRC	-	CRC according to NFC Forum	2 bytes
Data	-	Product version information	8 bytes
NAK	see Table 15	see Section 10.3	4 bit

Table 19. GET_VERSION timing

These times exclude the end of communication of the NFC device.

	T _{ACK/NAK} min	T _{ACK/NAK} max	T _{TimeOut}
GET_VERSION	n=9 ^[1]	T _{TimeOut}	5 ms

[1] Refer to [Section 10.2](#).

Table 20. GET_VERSION response for CRN120

Byte no.	Description	CRN120	Interpretation
0	fixed Header	00h	
1	vendor ID	04h	NXP Semiconductors
2	product type	04h	Charging receiver
3	product subtype	05h	50 pF; I ² C interface
4	major product version	02h	2
5	minor product version	02h	V2
6	storage size	13h	see following information
7	protocol type	03h	NFC Forum Type 2 Tag

The most significant 7 bits of the storage size byte are interpreted as an unsigned integer value n . As a result, it codes the total available user memory size as 2^n . If the least significant bit is 0b, the user memory size is exactly 2^n . If the least significant bit is 1b, the user memory size is between 2^n and 2^{n+1} .

10.6 PWD_AUTH

A protected memory area can be accessed only after a successful password verification using the PWD_AUTH command. The AUTH0 configuration byte defines the start of the protected area. It specifies the first page that the password mechanism protects. The level of protection can be configured using the NFC_PROT bit either for write protection or read/write protection. The PWD_AUTH command takes the password as parameter and, if successful, returns the password authentication acknowledge, PACK. By setting the AUTHLIM configuration bits to a value larger than 000b, the number of unsuccessful password verifications can be limited. Each unsuccessful authentication is then counted. After reaching the limit (2^{AUTHLIM}) of unsuccessful attempts, the memory write access or the memory access at all (specified in NFC_PROT) to the protected area, is no longer possible. The PWD_AUTH command is shown in [Figure 17](#) and [Table 21](#).

[Table 22](#) shows the required timing.

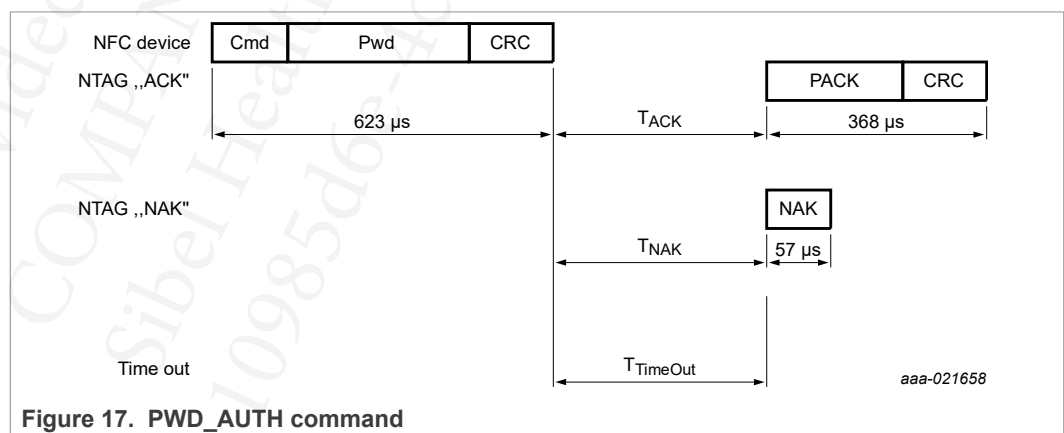


Table 21. PWD_AUTH command

Name	Code	Description	Length
Cmd	1Bh	password authentication	1 byte
Pwd	-	password	4 bytes
CRC	-	CRC according to NFC Forum	2 bytes
PACK	-	password authentication acknowledge	2 bytes
NAK	see Table 15	see Section 10.3	4-bit

Table 22. PWD_AUTH timing

These times exclude the end of communication of the NFC device.

	T _{ACK/NAK} min	T _{ACK/NAK} max	T _{TimeOut}
PWD_AUTH	n=9 ^[1]	T _{TimeOut}	5 ms

[1] Refer to [Section 10.2](#).

Remark: It is strongly recommended to change - and diversify for each tag - the password and PACK from its delivery state at tag issuing.

10.7 READ

The READ command requires a start page address, and returns the 16 bytes of four CRN120 pages. For example, if address (Addr) is 03h then pages 03h, 04h, 05h, 06h are returned. Special conditions apply if the READ command address is near the end of the accessible memory area. For details on those cases and the command structure, refer to [Figure 18](#) and [Table 23](#).

[Table 24](#) shows the required timing.

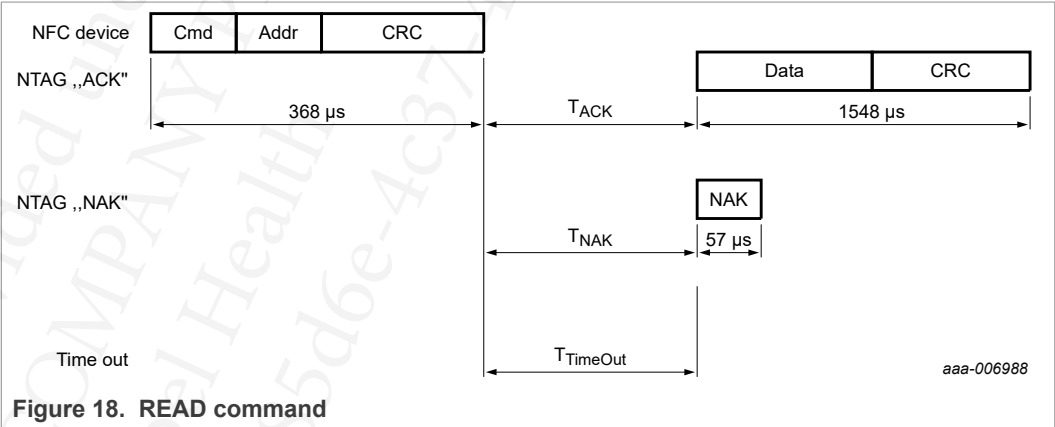


Table 23. READ command

Name	Code	Description	Length
Cmd	30h	read four pages	1 byte
Addr	-	start page address	1 byte

Table 23. READ command...continued

Name	Code	Description	Length
CRC	-	CRC according to NFC Forum	2 bytes
Data	-	Data content of the addressed pages	16 bytes
NAK	see Table 15	see Section 10.3	4 bit

Table 24. READ timing

These times exclude the end of communication of the NFC device.

	T _{ACK/NAK} min	T _{ACK/NAK} max	T _{TimeOut}
READ	n=9 ^[1]	T _{TimeOut}	5 ms

[1] Refer to [Section 10.2](#).

In the initial state of CRN120, all memory pages are allowed as Addr parameter to the READ command:

- Page address from 00h to E9h and pages ECh and EDh for CRN120
- SRAM buffer address when pass-through mode is enabled

Addressing a start memory page beyond the limits above results in a NAK response from CRN120.

In case a READ command addressing start with a valid memory area but extends over an invalid memory area, the content of the invalid memory area will be reported as 00h.

10.8 FAST_READ

The FAST_READ command requires a start page address and an end page address and returns all n*4 bytes of the addressed pages. For example, if the start address is 03h and the end address is 07h, then pages 03h, 04h, 05h, 06h and 07h are returned.

For details on those cases and the command structure, refer to [Figure 19](#) and [Table 25](#).

[Table 26](#) shows the required timing.

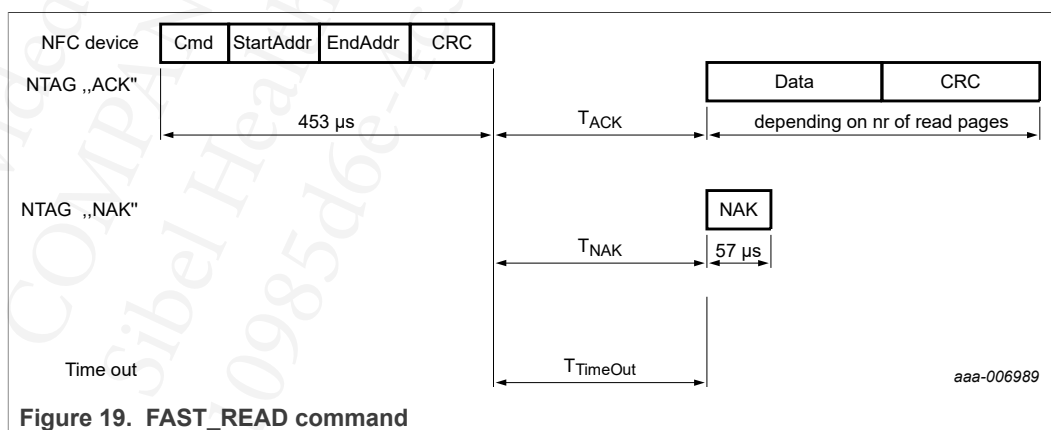


Figure 19. FAST_READ command

Table 25. FAST_READ command

Name	Code	Description	Length
Cmd	3Ah	read multiple pages	1 byte
StartAddr	-	start page address	1 byte
EndAddr	-	end page address	1 byte
CRC	-	CRC according to NFC Forum	2 bytes
Data	-	data content of the addressed pages	n*4 bytes
NAK	see Table 15	see Section 10.3	4 bit

Table 26. FAST_READ timing

These times exclude the end of communication of the NFC device.

	T _{ACK/NAK} min	T _{ACK/NAK} max	T _{TimeOut}
FAST_READ	n=9 ^[1]	T _{TimeOut}	5 ms

[1] Refer to [Section 10.2](#).

In the initial state of CRN120, all memory pages are allowed as StartAddr parameter to the FAST_READ command:

- Page address from 00h to E9h and pages ECh and EDh for CRN120
- SRAM buffer address when pass-through mode is enabled

If the start addressed memory page (StartAddr) is outside of accessible area, CRN120 replies a NAK.

In case the FAST_READ command starts with a valid memory area but extends over an invalid memory area, the content of the invalid memory area will be reported as 00h.

The EndAddr parameter must be equal to or higher than the StartAddr.

Remark: The FAST_READ command is able to read out the entire memory of one sector with one command. Nevertheless, the receive buffer of the NFC device must be able to handle the requested amount of data as no chaining is possible.

10.9 WRITE

The WRITE command requires a page address, and writes 4 bytes of data into the addressed CRN120 page. The WRITE command is shown in [Figure 20](#) and [Table 27](#).

[Table 28](#) shows the required timing.

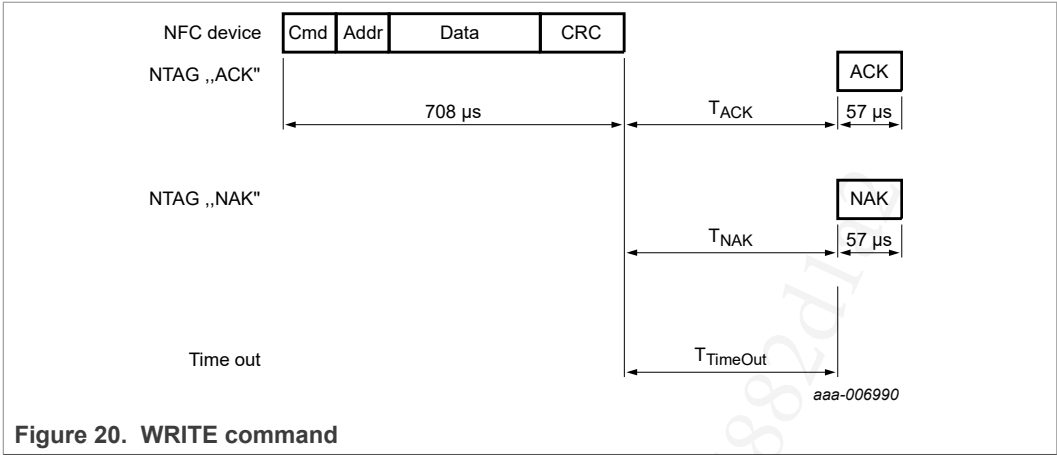


Table 27. WRITE command

Name	Code	Description	Length
Cmd	A2h	write one page	1 byte
Addr	-	page address	1 byte
Data	-	data	4 bytes
CRC	-	CRC according to NFC Forum	2 bytes
NAK	see Table 15	see Section 10.3	4 bit

Table 28. WRITE timing

These times exclude the end of communication of the NFC device.

	T _{ACK/NAK} min	T _{ACK/NAK} max	T _{TimeOut}
WRITE	n=9 ^[1]	T _{TimeOut}	5 ms

[1] Refer to [Section 10.2](#).

In the initial state of CRN120, the following memory pages are valid Addr parameters to the WRITE command:

- Page address from 02h to E9h (Sector 0) for CRN120
- SRAM buffer addresses when pass-through mode is enabled

Addressing a memory page beyond the limits above results in a NAK response from CRN120.

Pages that are locked against writing cannot be reprogrammed using any write command. The locking mechanisms include static and dynamic lock bits, as well as the locking of the configuration pages.

10.10 FAST_WRITE

The FAST_WRITE allows writing data in ACTIVE state to the complete SRAM (64 bytes) in pass-through mode. It requires the start block address (F0h), end address (FFh) and writes 64 bytes of data into the CRN120 SRAM. The FAST_WRITE command is shown in [Figure 21](#) and [Table 29](#).

WARNING: Data is written directly to SRAM. If received CRC at the end of transmission is wrong and response was a NAK, received (corrupted) data is still in SRAM. Hence it is recommended to implement a protocol on top to ensure data integrity (e.g. include own CRC at the end of the payload) when using SRAM.

Table 30 shows the required timing.

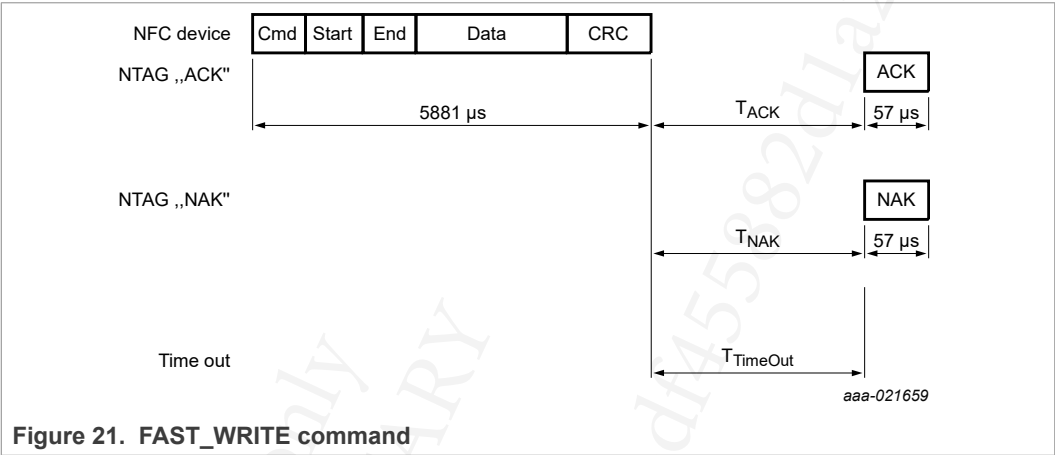


Figure 21. FAST_WRITE command

Table 29. FAST_WRITE command

Name	Code	Description	Length
Cmd	A6h	write complete SRAM	1 byte
START_ADDR	F0h	start SRAM in pass-through mode	1 byte
END_ADDR	FFh	end SRAM in pass-through mode	1 byte
Data	-	data	64 bytes
-	CRC	CRC according to NFC Forum	2 bytes
ACK	see Table 15	see Section 10.3	4 bit
NAK	see Table 15	see Section 10.3	4 bit

Table 30. FAST_WRITE timing

These times exclude the end of communication of the NFC device.

	T _{ACK/NAK min}	T _{ACK/NAK max}	T _{TimeOut}
FAST_WRITE	n=9 ^[1]	T _{TimeOut}	5 ms

[1] Refer to [Section 10.2](#).

11 Communication and arbitration between NFC and I²C interface

If both interfaces are powered by their corresponding source, only one interface shall have access to the memory according to the "first-come, first-serve" principle.

In NS_REG, the two status bits I2C_LOCKED and RF_LOCKED reflect the status of the CRN120 memory access and indicate which interface is locking the memory access. At power-on, both bits are 0b, setting the arbitration in idle mode.

In the case arbiter locks to the I²C interface, an NFC device can still read the session registers. If the NFC state machine is in ACTIVE state, only the SECTOR SELECT command is allowed. But any other command requiring EEPROM access like READ or WRITE is handled as an illegal command and replied to with a NAK value.

In the case where the memory access is locked to the NFC interface, the I²C host can still access the session register, by issuing a 'Register READ/WRITE' command. All other read or write commands will be replied to with a NACK to the I²C host.

11.1 Pass-through mode not activated

PTHRU_ON_OFF = 0b (see [Table 11](#)) indicates non-pass-through mode.

11.1.1 I²C interface access

If the tag is in the IDLE or HALT state (NFC state after POR or HALT-command) and the correct I²C slave address of CRN120 is received following the START condition, the bit I2C_LOCKED will be automatically set to 1b. If I2C_LOCKED = 1b, the I²C interface has access to the tag memory and the tag will respond with a NACK to any memory READ/WRITE command on the NFC interface other than reading the session register bytes during this time.

I2C_LOCKED must be either reset to 0b at the end of the I²C sequence or will be cleared automatically after the end of the watchdog timer.

11.1.2 NFC interface access

The arbitration allows the NFC interface read and write accesses to EEPROM only when I2C_LOCKED is set to 0b.

RF_LOCKED is automatically set to 1b if the tag receives a valid command (EEPROM access commands) on the NFC interface. If RF_LOCKED = 1b, the tag is locked to the NFC interface and will not respond to any command from the I²C interface other than READ register command (see [Table 11](#)).

RF_LOCKED is automatically set to 0b in one of the following conditions:

- At POR or if the NFC field is switched off
- If the tag is set to the HALT state with a HALT command on the NFC interface
- If the memory access command is finished on the NFC interface

When the NFC interface has read the last page of the NDEF message specified in LAST_NDEF_BLOCK (see [Table 9](#) and [Table 10](#)) the bit NDEF_DATA_READ - in the register NS_REG see [Table 11](#) - is set to 1b and indicates to the I²C interface that, for example, new NDEF data can be written.

11.2 SRAM buffer mapping with Memory Mirror enabled

With SRAM_MIRROR_ON_OFF= 1b, the SRAM buffer mirroring is enabled. This mode cannot be combined with the pass-through mode (see [Section 11.3](#)).

With the memory mirror enabled, the SRAM is now mapped into the user memory from the NFC interface perspective using the SRAM mirror lower page address specified in SRAM_MIRROR_BLOCK byte ([Table 9](#) and [Table 10](#)). See [Table 31](#) for an illustration of this SRAM memory mapping when SRAM_MIRROR_BLOCK is set to 01h.

Password protection to this mapped SRAM may be enabled by enabling password authentication and setting SRAM_PROT bit to 1b.

In contrary to password protection, for read only locking there are no special lock bits for the SRAM. Whenever user EEPROM blocks are locked to read-only with static and/or dynamic lock bits, potential mirrored SRAM blocks are read-only, too.

The tag must be VCC powered to make this mode work, because without VCC, the SRAM will not be accessible via NFC powered only.

When mapping the SRAM buffer to the user memory, the user shall be aware that all data written into the SRAM will be lost once the CRN120 is no longer powered from the I²C side (as SRAM is a volatile memory).

Table 31. Illustration of the SRAM memory addressing via the NFC interface (with SRAM_MIRROR_ON_OFF set to 1b and SRAM_MIRROR_BLOCK set to 01h) for the CRN120

Sector address	Page address		Byte number within a page				Access cond. ACTIVE state	Access cond. AUTH. state
	Dec.	Hex.	0	1	2	3		
0	0	00h	Serial number (UID)				READ	
	1	01h	Serial number (UID)			Internal	READ	
	2	02h	Internal		Static lock bytes		READ/R&W	
	3	03h	Capability Container (CC)				READ&WRITE	
	4	04h	SRAM				READ&WRITE	
	...	...						
	19	13h						
	...	...	Unprotected user memory				READ&WRITE	
	AUTH0	AUTH0	Protected user memory				READ	READ&WRITE
	...	...						
	225	E1h						
	226	E2h	Dynamic lock bytes			00h	R&W/READ	
	227	E3h	RFU	RFU	RFU	AUTH0	READ	READ&WRITE
	228	E4h	ACCESS	RFU	RFU	RFU	READ	READ&WRITE
	229	E5h	PWD				READ	READ&WRITE
	230	E6h	PACK		RFU	RFU	READ	READ&WRITE
	231	E7h	PT_I2C	RFU	RFU	RFU	READ	READ&WRITE
232	E8h	Configuration registers				see Table 9		
233	E9h							

Table 31. Illustration of the SRAM memory addressing via the NFC interface (with SRAM_MIRROR_ON_OFF set to 1b and SRAM_MIRROR_BLOCK set to 01h) for the CRN120...continued

Sector address	Page address		Byte number within a page				Access cond. ACTIVE state	Access cond. AUTH. state
	Dec.	Hex.	0	1	2	3		
	234	EAh	Invalid access - returns NAK				n.a.	
	235	EBh						
	236	ECh	Session registers				see Table 10	
	237	EDh						
	238	EEh	Invalid access - returns NAK				n.a.	
	239	EFh						
	240	F0h	Invalid access - returns NAK				n.a.	
	...	...						
	255	FFh						

11.3 Pass-through mode

PTHRU_ON_OFF = 1b (see [Table 11](#)) enables and indicates pass-through mode.

Password protection for pass-through mode may be enabled by enabling password authentication and setting SRAM_PROT bit to 1b.

To handle large amount of data transfer from one interface to the other, CRN120 offers the pass-through mode where data is transferred via a 64 byte SRAM. This buffer offers fast write access and unlimited write endurance as well as an easy handshake mechanism between the two interfaces.

This buffer is mapped directly at the end of the Sector 0 of CRN120.

In both directions, the principle of access to the SRAM buffer via the NFC and I²C interface is exactly the same (see [Section 11.3.2](#) and [Section 11.3.3](#)).

The data flow direction must be set with the TRANSFER_DIR bit (see [Table 12](#)) within the current communication session using the session registers (it can only be set via the I²C interfaces) or for the configuration bits after POR (in this case both NFC and I²C interface can set it). This pass-through direction setting avoids locking the memory access during the data transfer from one interface to the SRAM buffer.

The pass-through mode can only be enabled via I²C interface when both interfaces are powered. The PTHR_U_ON_OFF bit, located in the session registers NC_REG (see [Table 12](#)), needs to be set to 1b. In case one interface powers off, the pass-through mode is disabled automatically.

CRN120 introduces in addition to the FAST_READ command a FAST_WRITE command. With this new command in ACTIVE state whole SRAM can be written at once, which improves the total pass-through performance significantly.

11.3.1 SRAM buffer mapping

In pass-through mode, the SRAM of CRN120 is mirrored to pages F0h to FFh of Sector 0.

The last page/block of the SRAM (page FFh) is used as the terminator page. Once the terminator page/block in the respective interfaces is read/written, the control would be

transferred to other interface (NFC/I²C) - see [Section 11.3.2](#) and [Section 11.3.3](#) for more details.

Accordingly, the application can align on the reader and host side to transfer 16/32/48/64 bytes of data in one pass-through step by only using the last blocks/page of the SRAM buffer.

For best performance in addition to the FAST_READ, the FAST_WRITE command should be used.

Table 32. Illustration of the SRAM memory addressing via the NFC interface in pass-through mode (PTHRU_ON_OFF set to 1b) for the CRN120

Sector address	Page address		Byte number within a page				Access cond.	Access cond.
	Dec.	Hex.	0	1	2	3	ACTIVE state	AUTH. state
0	0	00h	Serial number (UID)				READ	
	1	01h	Serial number (UID)			Internal	READ	
	2	02h	Internal		Static lock bytes		READ/R&W	
	3	03h	Capability Container (CC)				READ&WRITE	
	4	04h	Unprotected user memory				READ&WRITE	
	...	...						
	AUTH0	AUTH0	Protected user memory				READ	READ&WRITE
	...	...						
	225	E1h						
	226	E2h	Dynamic lock bytes			00h	R&W/READ	
	227	E3h	RFU	RFU	RFU	AUTH0	READ	READ&WRITE
	228	E4h	ACCESS	RFU	RFU	RFU	READ	READ&WRITE
	229	E5h	PWD				READ	READ&WRITE
	230	E6h	PACK		RFU	RFU	READ	READ&WRITE
	231	E7h	PT_I2C	RFU	RFU	RFU	READ	READ&WRITE
	232	E8h	Configuration registers				see Table 9	
	233	E9h						
	234	EAh	Invalid access - returns NAK				n.a.	
	235	EBh						
	236	ECh	Session registers				see Table 10	
	237	EDh						
	238	EEh	Invalid access - returns NAK				n.a.	
	239	EFh						
	240	F0h	SRAM				READ&WRITE	
	...	...						
255	FFh							

11.3.2 NFC to I²C data transfer

If the NFC interface is enabled (RF_LOCKED = 1b) and data is written to the terminator page FFh of the SRAM via the NFC interface, at the end of the WRITE command, bit SRAM_I2C_READY is set to 1b and bit RF_LOCKED is set to 0b automatically, and the CRN120 is locked to the I²C interface.

To signal the host that data is ready to be read following mechanisms are in place:

- The host polls/reads bit SRAM_I2C_READY from NS_REG (see [Table 12](#)) to know if data is ready in SRAM
- A trigger on the ED pin indicates to the host that data is ready to be read from SRAM. This feature can be enabled by programming bits 5:2 (ED_OFF, ED_ON) of the NC_REG appropriately (see [Table 11](#))

This is illustrated in the [Figure 22](#).

If the tag is addressed with the correct I²C slave address, the I2C_LOCKED bit is automatically set to 1b (according to the interface arbitration). After a READ from the terminator page of the SRAM, bit SRAM_I2C_READY and bit I2C_LOCKED are automatically reset to 0b. The tag returns to the arbitration idle mode where, for example, further data from the NFC interface can be transferred.

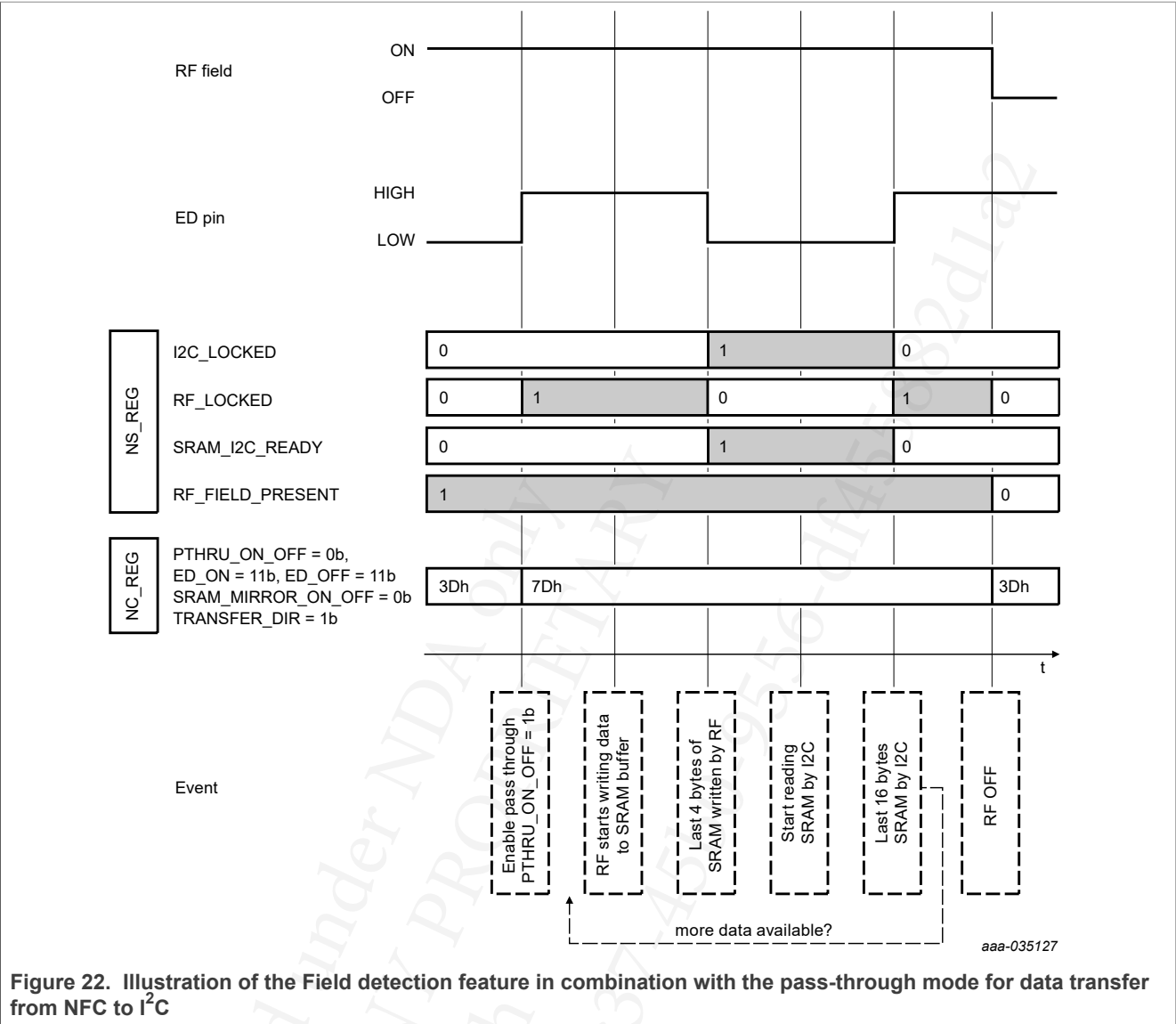


Figure 22. Illustration of the Field detection feature in combination with the pass-through mode for data transfer from NFC to I²C

11.3.3 I²C to NFC data transfer

If the I²C interface is enabled (I2C_LOCKED is 1b) and data is written to the terminator block FBh of the SRAM via the I²C interface, at the end of the WRITE command, bit SRAM_RF_READY is set to 1b and bit I2C_LOCKED is automatically reset to 0b to set the tag in the arbitration idle state.

The RF_LOCKED bit is then automatically set to 1b (according to the interface arbitration). After a READ or FAST_READ command involving the terminator page of the SRAM, bit SRAM_RF_READY and bit RF_LOCKED are automatically reset to 0b allowing the I²C interface to further write data into the SRAM buffer.

To signal to the host that further data is ready to be written, the following mechanisms are in place:

- The NFC interface polls/reads the bit SRAM_RF_READY from NS_REG (see Table 12) to know if new data has been written by the I²C interface in the SRAM

- A trigger on the ED pin indicates to the host that data has been read from SRAM by the NFC interface. This feature can be enabled by programming bits 5:2 (ED_OFF, ED_ON) of the NC_REG appropriately (see [Table 11](#))

The above mechanism is illustrated in the [Figure 23](#).

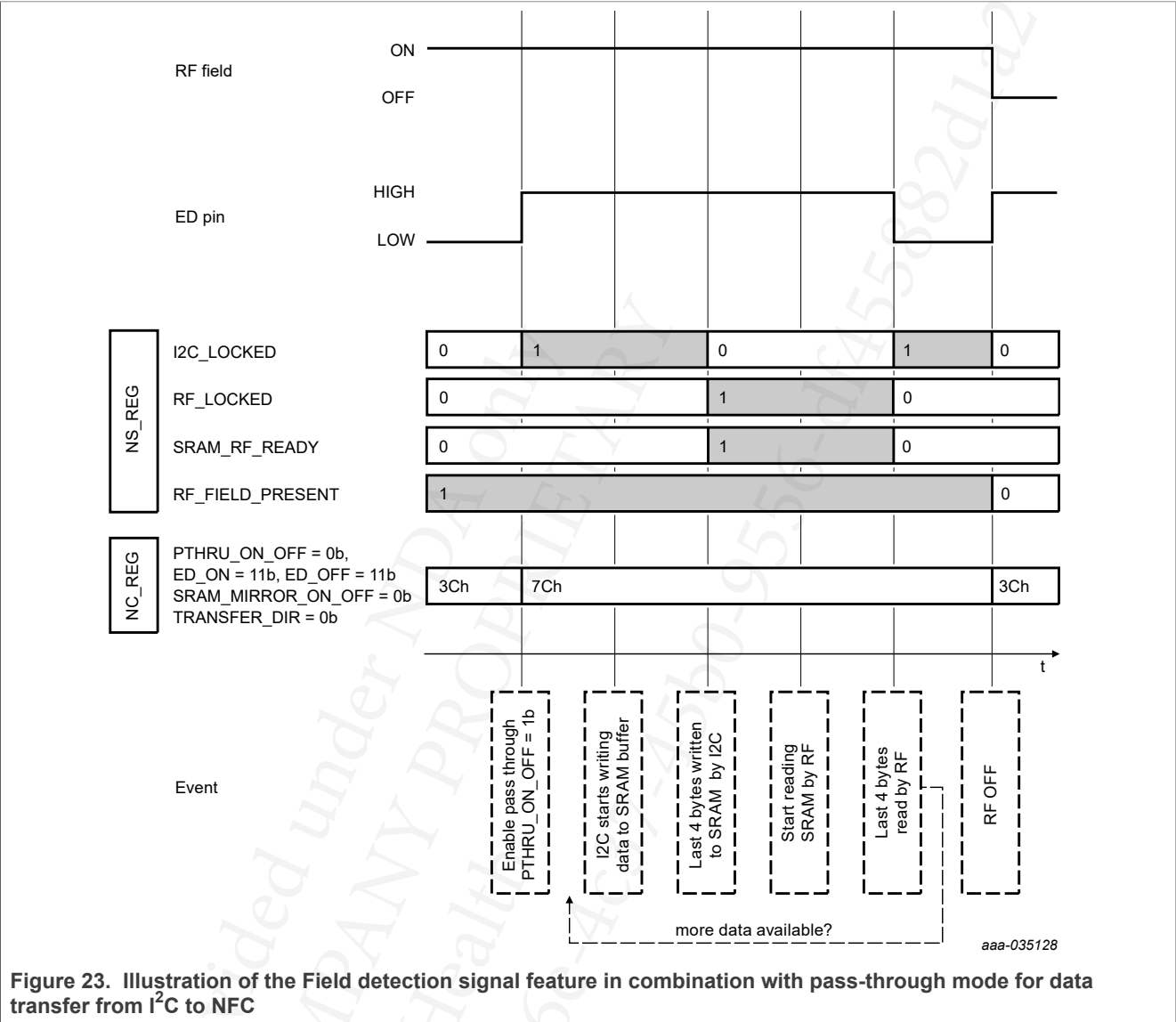


Figure 23. Illustration of the Field detection signal feature in combination with pass-through mode for data transfer from I²C to NFC

12 Limiting values

Exceeding the limits of one or more values in reference may cause permanent damage to the device. Exposure to limiting values for extended periods may affect device reliability.

Table 33. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134). [1] [2]

Symbol	Parameter	Conditions	Min	Max	Unit
T_{stg}	storage temperature		-55	+125	°C
$T_{\text{j(max)}}$	maximum junction temperature		-	+105	°C
V_{ESD}	electrostatic discharge voltage	human body model (HBM) ^[3]	-	2	kV
		charge device model (CDM) ^[4]	-	1	kV
V_{DD}	supply voltage	on pin VCC	-0.5	4.6	V
V_{i}	input voltage	on pin ED, SDA, SCL	-0.5	4.6	V
I_{i}	input current	on pin LA, LB	-	40	mA
$V_{\text{i(RF)}}$	RF input voltage	on pin LA, LB	-	4.6	V_{peak}

[1] Stresses above one or more of the limiting values may cause permanent damage to the device.

[2] Exposure to limiting values for extended periods may affect device reliability.

[3] According to ANSI/ESDA/JEDEC JS-001.

[4] According to ANSI/ESDA/JEDEC JS-002.

13 Characteristics

13.1 Electrical characteristics

Table 34. Characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
C _i	input capacitance	LA - LB, on-chip - C _{IC} , f=13.56 MHz, V _{LA-LB} =2.4 V _{RMS}	44	50	56	pF
f _i	input frequency		-	13.56	-	MHz
T _{amb}	operating ambient temperature		-40	25	+85	°C
R _{TH_JA}	thermal resistance	JEDEC 2s2p board and XQFN8 package	-	150	-	K/W
Field detect pin						
V _{FD}	output voltage	4.7 nF; 40 kΩ	-	-	3.6	V
V _{OH}	HIGH-level output voltage	NFC field on; 4.7 nF; 40 kΩ; load current less than 150 μA	2.4	-	-	V
V _{OL}	LOW-level output voltage	NFC field off; 4.7 nF; 40 kΩ	-	-	0.7	V
I _{FD}	Maximum drive capability	4.7 nF; 40 kΩ	-	-	150	μA
I ² C interface characteristics						
V _{CC}	supply voltage	supplied via V _{CC} only	1.67	-	3.6	V
I _{DD}	supply current	V _{CC} =1.8 V I ² C; idle bus	-	160	-	μA
		V _{CC} =3.3 V I ² C; idle bus	-	195	-	μA
I _{DD}	supply current	V _{CC} =1.8 V I ² C@400KHz	-	-	185	μA
		V _{CC} =2.5 V I ² C@400KHz	-	-	210	μA
		V _{CC} =3.3 V I ² C@400KHz	-	-	240	μA
I ² C pin characteristics						
V _{OL}	LOW-level output voltage	I _{OL} = 3 mA; V _{CC} > 2 V	-	-	0.4	V
		I _{OL} = 2 mA; V _{CC} < 2 V	-	-	0.2*V _{CC}	V
V _{IH}	HIGH-level input voltage		0.7*V _{CC}	-	-	V
V _{IL}	LOW-level input voltage		-	-	0.3*V _{CC}	V
C _i	input capacitance	SCL and SDA pin	-	2.4	-	pF
I _L	leakage current	0 V and V _{CC,max}	-	-	10	μA
t _{high}	SCL high time	fast mode 400 kHz	950	-	-	ns
ED pin characteristics						
V _{OL}	LOW-level output voltage	I _{OL} = 4 mA; V _{CC} > 2 V	-	-	0.4	V
		I _{OL} = 3 mA; V _{CC} < 2 V	-	-	0.2*V _{CC}	V
I _L	leakage current		-	1.5	10	μA
EEPROM characteristics						